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Chapter 1

Linear Algebra and Regression

1.1 Review of statistical concepts

1.1.1 OLS

There are two core assumptions in OLS:

Assumption 1.1.1: A1

Linearity: $y_t = \beta_1 + \beta_2 x_t + u_t$.

If A1 fails, use non-parametric.

Assumption 1.1.2: A2

Exogeneity: $E(u_t \,|\, x_t) = 0$

From A2, we know

$$E(y_t \,|\, x_t) = \beta_1 + \beta_2 x_t + E(u_t \,|\, x_t) = \beta_1 + \beta_2 x_t$$

If A2 fails, we have endogeneity and we could fix with instrumental-variable (IV) estimation.
What if we care about the whole distribution, not just the mean? Quantile regression shows how x shifts the median, upper tail, lower tail, etc., revealing heterogeneity across the outcome distribution.

1.1.2 Misspecification

We start with a normal OLS

$$y_t = \beta_1 + \beta_2 x_t + u_t, \quad E(u_t \mid x_t) = 0, \quad E(y_t \mid x_t) = \beta_1 + \beta_2 x_t$$

But there is an omitted quadratic term, so the true model is

$$y_t = \beta_1 + \beta_2 x_t \; + \; \underbrace{\left(\beta_3 x_t^2 + v_t\right)}_{u_t}$$

The omitted piece gets absorbed into the error

$$u_t = \beta_3 x_t^2 + v_t$$

Therefore

$$E(u_t \mid x_t) = \beta_3 x_t^2 + E(v_t \mid x_t) = \beta_3 x_t^2 \neq 0$$

Exogeneity fails and then OLS is biased and inconsistent. We call this Misspecification.

1.1.3 Log-linear Form

Definition 1.1.1

$$y_t = x_t^{\beta_2} \Rightarrow \log y_t = \beta_2 \log x_t \Rightarrow y_t = \beta_2 x_t$$

We call it log-linear form.

If there is a constant term and random noise

$$y_t = e^{\beta_1} x_t^{\beta_2} + u_t \Rightarrow \log y_t = \log\big(e^{\beta_1} x_t^{\beta_2} + u_t\big)$$

$\log\big(e^{\beta_1} x_t^{\beta_2} + u_t\big)$ cannot be linearized. So instead,

$$y_t = e^{\beta_1} x_t^{\beta_2} e^{u_t} \Rightarrow \log y_t = \log\big(e^{\beta_1} x_t^{\beta_2} e^{u_t}\big) = \beta_1 + \beta_2 \log x_t + u_t \Rightarrow y_t = \alpha_1 + \alpha_2 x_t + v_t$$

1.2 Review of Linear Algebra

1.2.1 Vector Operations

Definition 1.2.1

$\mathbb{R}^m$ is real vector space.

Example.

$$\mathbf{x} = \begin{bmatrix} 5 \\ 7 \\ 10 \\ 22 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

When $m = 1$, it is a single-entry vector reduces to a scalar. There are two special vectors:

Definition 1.2.2

$$\mathbf{O}_m = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{1}_m = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \mathbf{i}_m$$

Definition 1.2.3

$\forall\, \mathbf{x}, \mathbf{y} \in \mathbb{R}^m$,

- $\mathbf{x} = \mathbf{y} \iff x_i = y_i \quad \forall i = 1, \ldots, m$
- $\mathbf{x} > \mathbf{y} \iff x_i > y_i \quad \forall i = 1, \ldots, m$
- $\mathbf{x} \geq \mathbf{y} \iff x_i \geq y_i \quad \forall i = 1, \ldots, m$

Example.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \implies \mathbf{x} \neq \mathbf{y},\ \mathbf{x} \leq \mathbf{y}$$

Definition 1.2.4: Addition

$\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$,

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_m + y_m \end{bmatrix}$$

Example.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \mathbf{x} + \mathbf{y} = \begin{bmatrix} 1+4 \\ 2+5 \\ 3+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Definition 1.2.5: Multiplication by a scalar

$$\lambda \in \mathbb{R}, \ \lambda \mathbf{x} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_m \end{bmatrix}$$

Example.

$$\lambda = 2, \ 2\mathbf{x} = \begin{bmatrix} 2 \cdot 1 \\ 2 \cdot 2 \\ 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Definition 1.2.6

$\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$ are colinear if $\mathbf{x} = \lambda \mathbf{y} \quad$ for some $\lambda \in \mathbb{R}$

Example.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix} \Rightarrow \mathbf{y} = 4\mathbf{x}, \quad \lambda = 4.$$

Definition 1.2.7

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$. The inner product is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle \ = \ \sum_{i=1}^{m} x_i \, y_i.$$

Example.

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$

Then

$$\langle \mathbf{x}, \mathbf{y} \rangle = 1 \cdot 4 \ + \ 2 \cdot 5 \ + \ 3 \cdot 6 = 4 + 10 + 18 = 32.$$

Equivalently,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}^\top \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 32.$$

Definition 1.2.8

If $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, then $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Definition 1.2.9: Euclidean norm

$\langle \mathbf{x}, \mathbf{x} \rangle^{1/2} \ = \ \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} \ = \ \|\mathbf{x}\|$

Example.

$\mathbf{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ then $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{9+4} = \sqrt{13}$.

Definition 1.2.10: Unit vectors (normalization)

$\|\mathbf{x}\| = 1, \quad \|\mathbf{y}\| = 1$

Definition 1.2.11: Orthonormality

$\mathbf{x}, \mathbf{y}$ are unit vectors and $\langle \mathbf{x}, \mathbf{y} \rangle = 0 \implies \mathbf{x}$ and $\mathbf{y}$ are orthonormal.

Example.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\|\mathbf{e}_1\| = \sqrt{\langle \mathbf{e}_1, \mathbf{e}_1 \rangle} = \sqrt{1+0} = 1, \qquad \|\mathbf{e}_2\| = \sqrt{\langle \mathbf{e}_2, \mathbf{e}_2 \rangle} = \sqrt{0+1} = 1$ and $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = 1 \cdot 0 + 0 \cdot 1 = 0$.
Thus, there are orthonormal.

1.2.2 Matrix

Matrix Operations

Definition 1.2.12: Element-wise representation

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix} = [a_{ij}] = (a_{ij}), \quad i, j = 1, \dots, m.$$

$\mathbf{a}_{\cdot j}$: j-th column vector, $\quad \mathbf{a}_{i\cdot}$: i-th row vector.

Example.

Suppose we have two different 2×3 matrices:

$$\mathbf{A}_{2\times 3} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \neq \mathbf{B}_{2\times 3} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \end{pmatrix}.$$

Definition 1.2.13

$\mathbf{A} = \mathbf{B} \iff a_{ij} = b_{ij} \quad \forall\, i, j.$

Definition 1.2.14: Transpose

The transpose flips rows and columns:

$$\mathbf{A}^\top = \mathbf{A}' \in \mathbb{R}^{n\times m}, \qquad \left(\mathbf{A}^\top\right)_{ji} = a_{ij}.$$

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2\times 3} \implies \mathbf{A}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

Definition 1.2.15: Addition

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m\times m}$.

$$\mathbf{A} + \mathbf{B} \;=\; \left(a_{ij} + b_{ij}\right)_{i=1,\ldots,m;\; j=1,\ldots,m}.$$

Definition 1.2.16: Scalar multiplication

For a scalar $\lambda \in \mathbb{R}$,

$$\lambda\mathbf{A} \;=\; \left[\lambda\, a_{ij}\right]_{i,j}.$$

Example.

Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$

Then

$$\mathbf{A} + \mathbf{B} \;=\; \begin{pmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix} = 2\begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}.$$

Definition 1.2.17

Let $\mathbf{A} \in \mathbb{R}^{m\times m}$ and $\mathbf{B} \in \mathbb{R}^{m\times s}$. The product is defined when the inner dimensions match:

$$\mathbf{AB} = \left(c_{ik}\right)_{i=1,\ldots,m;\; k=1,\ldots,s}, \qquad c_{ik} = \sum_{j=1}^{m} a_{ij}\, b_{jk}.$$

Theorem 1.2.18

In general $\mathbf{AB} \neq \mathbf{BA}$.

Example.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

$$\mathbf{AB} = \begin{pmatrix} 1\cdot 5 + 2\cdot 7 & 1\cdot 6 + 2\cdot 8 \\ 3\cdot 5 + 4\cdot 7 & 3\cdot 6 + 4\cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

$$\mathbf{BA} = \begin{pmatrix} 5\cdot 1 + 6\cdot 3 & 5\cdot 2 + 6\cdot 4 \\ 7\cdot 1 + 8\cdot 3 & 7\cdot 2 + 8\cdot 4 \end{pmatrix} = \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}.$$

Thus $\mathbf{AB} \neq \mathbf{BA}$.

Theorem 1.2.19

$(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top \neq \mathbf{A}^\top\mathbf{B}^\top$

$(\mathbf{ABC})^\top = \mathbf{C}^\top\mathbf{B}^\top\mathbf{A}^\top$

Example.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{A}^\top = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}, \quad \mathbf{B}^\top = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

From previous example, we already know

$$\mathbf{AB} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

Then

$$(\mathbf{AB})^\top = \begin{pmatrix} 19 & 43 \\ 22 & 50 \end{pmatrix}$$

Then we know $\mathbf{B}^\top\mathbf{A}^\top = (\mathbf{AB})^\top$. Indeed,

$$\mathbf{B}^\top\mathbf{A}^\top = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 19 & 43 \\ 22 & 50 \end{pmatrix} = (\mathbf{AB})^\top$$

Types of Matrix

Definition 1.2.20

A square matrix $\mathbf{A}_{n\times n}$ has the same number of rows and columns

Example.

$$\mathbf{A}_{3\times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Definition 1.2.21

Diagonal matrix: every off-diagonal entry is zero

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Definition 1.2.22

Identity matrix: a diagonal matrix whose diagonal elements are all ones, denoted by $\mathbf{I}_m$.

Definition 1.2.23

Symmetric matrix: a square matrix that equals its own transpose $\mathbf{A}^\top = \mathbf{A}$.

It is obvious that every diagonal matrix is also a symmetric matrix.

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 2 \\ 3 & 2 & 4 \end{bmatrix} = \mathbf{A}^\top$$

Theorem 1.2.24

- $\mathbf{I}_m \mathbf{I}_m = \mathbf{I}_m$
- $\mathbf{A}\mathbf{I}_m = \mathbf{I}_m\mathbf{A} = \mathbf{A}$

Example.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{bmatrix} 1+0 & 0+2 \\ 3+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Definition 1.2.25: Trace

$\mathrm{tr}(\mathbf{A}) = \sum_i a_{ii}$

Example.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \mathrm{tr}(\mathbf{A}) = 5$$

Example.

$\mathrm{tr}(\mathbf{I}_m) = m.$

Definition 1.2.26

Matrix powers: if $\mathbf{A}$ is an $m \times m$ square matrix, the second power is simply the matrix multiplied by itself $\mathbf{A}^2 = \mathbf{A}\mathbf{A}$, which is also an $m \times m$ matrix.

Definition 1.2.27

Idempotent matrix: A special case where the square of the matrix returns the matrix $\mathbf{A}^2 = \mathbf{A}$.

Example.

$$\mathbf{A} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \mathbf{AA}$$

Definition 1.2.28

A square matrix $\mathbf{A}_{m\times m}$ is invertible when there exists a matrix $\mathbf{A}^{-1}{}_{m\times m}$ such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_m$.

Not all matrix are invertible.

Theorem 1.2.29

For a diagonal matrix the inverse is the reciprocals of the diagonal entries.

Example.

Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1.3 Matrix Form of OLS

Suppose we have $y_t = \beta_1 + \beta_2\, x_t + U_t$ and we want to transform this into a matrix form. For $t = 1, 2, \ldots, m$:

$$\begin{cases} t = 1: & y_1 & = 1 \cdot \beta_1 + \beta_2\, x_1 + U_1 \\ t = 2: & y_2 & = 1 \cdot \beta_1 + \beta_2\, x_2 + U_2 \\ & & \vdots \\ t = m: & y_m & = 1 \cdot \beta_1 + \beta_2\, x_m + U_m \end{cases}$$

We define

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}_{m\times 1}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}_{m\times 2}, \quad \mathbf{U} = \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_m \end{bmatrix}_{m\times 1}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}_{2\times 1}.$$

Then

$$\mathbf{X}_{m\times 2}\boldsymbol{\beta}_{2\times 1} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \beta_1 + x_1\beta_2 \\ \beta_1 + x_2\beta_2 \\ \vdots \\ \beta_1 + x_m\beta_2 \end{bmatrix}_{m\times 1}$$

Thus,

$$\underbrace{\mathbf{y}}_{m\times 1} = \underbrace{\mathbf{X}\boldsymbol{\beta}}_{m\times 1} + \underbrace{\mathbf{U}}_{m\times 1}$$

Suppose now we have two inputs: $y_t = \beta_1 + \beta_2 x_{1t} + \beta_3 x_{2t} + U_t$. Then

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{21} \\ 1 & x_{12} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{1m} & x_{2m} \end{bmatrix}_{m\times 3}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}_{3\times 1}$$

In general, we have $\mathbf{X}_{m\times K}$, which means m obversations and K inputs. The underline assumption is that the number of observations m greatly exceeds the number of parameters K for $\mathbf{X}_{m\times K}$.

1.4 Method of Moment

A very natural way to estimate parameters is to replace population means by sample means. This technique is called **the method of moments**, and it is one of the most widely-used estimation methods in statistics. Keep in mind that we have two assumptions: linearity and exogeneity.

1.4.1 MM for Intercept-only Model

We start with the simpliest case: $y_t = \beta_1 + u_t$:

1. Population moment condition: $E(u_t) = 0$ from the exogeneity condition $E(u_t|x_t) = 0$.
2. Replace u_t by its sample $\hat{u}_t = y_t - \hat{\beta}_1$.
3. Impose the sample moment condition $\frac{1}{m}\sum_t \hat{u}_t = 0$.
4. Expand:

$$\frac{1}{m}\sum_t \hat{u}_t = \frac{1}{m}\sum_t \big(y_t - \hat{\beta}_1\big) = 0 \Leftrightarrow \frac{1}{m}\sum_t y_t - \frac{1}{m}\sum_t \hat{\beta}_1 = 0 \Leftrightarrow \hat{\beta}_1 = \frac{1}{m}\sum_t y_t$$

since $\sum_t \hat{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_1 + \cdots + \hat{\beta}_1 = m\hat{\beta}_1$

The same value could be obtained by minimising the sum of squared residuals $\min_{\hat{\beta}_1} \sum_t \big(y_t - \hat{\beta}_1\big)^2$.

1.4.2 MM with One Regressor

Now we have one input: $y_t = \beta_1 + \beta_2\, x_t + u_t$. In this case, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$, where $\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}_{m\times 2}$.

By exogeneity $E\big(u_t \mid x_t\big) = 0 \;\Rightarrow\; E(u_t) = 0$. Rewrite $u_t = y_t - \beta_1 - \beta_2\, x_t$. Thus,

$$\frac{1}{m}\sum_t \hat{u}_t = 0 \Rightarrow \frac{1}{m}\sum_t \big(y_t - \hat{\beta}_1 - \hat{\beta}_2\, x_t\big) = 0 \tag{1.1}$$

We also have, by law of iterated expectations,

$$E\big(x_t u_t\big) = E_x\Big[\, E\big(x_t u_t \mid x_t\big)\Big] = E_x\big[x_t E\big(u_t \mid x_t\big)\big] = 0$$

Then

$$\frac{1}{m}\sum_t x_t \hat{u}_t = \frac{1}{m}\sum_t x_t\big(y_t - \hat{\beta}_1 - \hat{\beta}_2\, x_t\big) = 0 \tag{1.2}$$

Fact 1.4.1

- $\sum_t x_t u_t = x_1 u_1 + x_2 u_2 + \cdots + x_m u_m$
- $\sum_t (x_t + u_t) = x_1 + u_1 + x_2 + u_2 + \cdots + x_m + u_m = \sum_t x_t + \sum_t u_t$

Note, $\sum_t x_t u_t \neq \Big(\sum_t x_t\Big)\Big(\sum_t u_t\Big)$

By (1.1) and (1.2), we have

$$\begin{cases} E(u_t) = 0 \implies \frac{1}{m}\sum_t \big(y_t - \hat{\beta}_1 - \hat{\beta}_2 x_t\big) = 0 \\ E(x_t u_t) = 0 \implies \frac{1}{m}\sum_t x_t\big(y_t - \hat{\beta}_1 - \hat{\beta}_2 x_t\big) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{1}{m}\sum_t y_t \;=\; \hat{\beta}_1 \;+\; \hat{\beta}_2 \frac{1}{m}\sum_t x_t, \\ \frac{1}{m}\sum_t x_t y_t \;=\; \hat{\beta}_1 \frac{1}{m}\sum_t x_t \;+\; \hat{\beta}_2 \frac{1}{m}\sum_t x_t^2. \end{cases}$$

$$\Rightarrow \begin{cases} m\hat{\beta}_1 + \hat{\beta}_2 \sum_t x_t = \sum_t y_t, \\ \hat{\beta}_1 \sum_t x_t + \hat{\beta}_2 \sum_t x_t^2 = \sum_t x_t y_t. \end{cases}$$

$$\Rightarrow \begin{bmatrix} m & \sum_t x_t \\ \sum_t x_t & \sum_t x_t^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} \sum_t y_t \\ \sum_t x_t y_t \end{bmatrix} \tag{1.3}$$

Recall.

$$\mathbf{1}_m = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}.$$

Then

$$\mathbf{1}_m^\top \mathbf{y} = y_1 + y_2 + \cdots + y_m = \sum_t y_t, \quad \mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_m y_m = \sum_t x_t y_t.$$

Then

$$\begin{bmatrix} \mathbf{1}_m^\top \\ \mathbf{x}^\top \end{bmatrix} \mathbf{y} = \begin{bmatrix} \sum_t y_t \\ \sum_t x_t y_t \end{bmatrix}, \qquad \begin{bmatrix} \mathbf{1}_m^\top \\ \mathbf{x}^\top \end{bmatrix} \begin{bmatrix} \mathbf{1}_m & \mathbf{x} \end{bmatrix} = \begin{bmatrix} m & \sum_t x_t \\ \sum_t x_t & \sum_t x_t^2 \end{bmatrix}.$$

From the RHS of (1.3),

$$\begin{bmatrix} \sum_t y_t \\ \sum_t x_t y_t \end{bmatrix} = \begin{bmatrix} \mathbf{1}_m^\top \mathbf{y} \\ \mathbf{x}^\top \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{1}_m^\top \\ \mathbf{x}^\top \end{bmatrix} \mathbf{y} = \mathbf{X}^\top \mathbf{y}$$

From the LHS of (1.3),

$$\begin{bmatrix} m & \sum_t x_t \\ \sum_t x_t & \sum_t x_t^2 \end{bmatrix} = \begin{bmatrix} \mathbf{1}_m^\top \mathbf{1}_m & \mathbf{1}_m^\top \mathbf{x} \\ \mathbf{x}^\top \mathbf{1}_m & \mathbf{x}^\top \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{1}_m^\top \\ \mathbf{x}^\top \end{bmatrix} \begin{bmatrix} \mathbf{1}_m & \mathbf{x} \end{bmatrix} = \mathbf{X}^\top \mathbf{X}$$

Then the ordinary-least-squares estimator (it's interesting that it's not called MM estimator) is

$$(\mathbf{X}^\top \mathbf{X})\, \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y} \qquad \Longrightarrow \qquad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \tag{2.4}$$

Provided $\mathbf{X}^\top \mathbf{X}$ is invertible ($\det(\mathbf{X}^\top \mathbf{X}) \neq 0$). Equivalently, it leads to our A3:

Assumption 1.4.1: A3

$\mathbf{X}$ is full-rank, which is equivalent to that there is no perfect multicollinearity (all columns are linear independent).

We can do Auxiliary Regressions to test A3:
Given $\mathbf{X} = \begin{bmatrix} \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k \end{bmatrix}$, we use $\mathbf{X}_{-j} = \begin{bmatrix} \mathbf{x}_1, \ldots, \mathbf{x}_{j-1}, \mathbf{x}_{j+1}, \ldots, \mathbf{x}_k \end{bmatrix} \quad \forall j$ to regress

$$\mathbf{x}_j = \mathbf{X}_{-j}\, \delta + \mathbf{v} \quad \forall j.$$

Then we test $\sum_t \hat{\mathbf{v}}^{\mathbf{2}} = 0$.

For the simple regression with a single regressor, (2.4) reduces to

$$\hat{\beta}_2 = \frac{\mathrm{Cov}(\mathbf{x}, \mathbf{y})}{\mathrm{Var}(\mathbf{x})}$$

1.4.3 MM for K Regressors

In general, for a multiple-regression model with K regressors and m observations,

$$y_t = \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_K x_{Kt} + u_t \Leftrightarrow \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_K)$.
By exogeneity, $E\big(u_t \mid x_{it}\big) = 0 \quad \text{for } i = 1, \ldots, K \Rightarrow E\big(x_{it} u_t\big) = 0 \quad (K \text{ population moment conditions})$.
Replace u_t by the residual $\hat{u}_t = y_t - \mathbf{x}_t^\top \hat{\boldsymbol{\beta}}$, and impose the same conditions on the sample

$$\frac{1}{m} \sum_t x_{it} \hat{u}_t = 0 \Rightarrow \sum_t x_{it} \big(y_t - \mathbf{x}_t^\top \hat{\boldsymbol{\beta}}\big) = \sum_t x_{it} \big((\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})_t\big) = \mathbf{x}_i^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0 \quad \text{for } i = 1, \ldots, K$$

Stacking these K equations gives

$$\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0 \implies \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = 0 \implies \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

In summary, model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad m > K$$

There are three assumptions:

- Linearity in parameters.
- Exogeneity: $E(u_t \mid x_t) = 0$ $\big(E(x_{it} u_t) = 0 \ \forall i = 1, \ldots, K\big)$. Note that there is a strong version $E(\mathbf{u} \mid \mathbf{X}) = 0$.
- Full rank of $\mathbf{X}$: $\mathbf{X}^\top \mathbf{X}$ is invertible (no perfect multicollinearity)

Theorem 1.4.2: Ordinary-Least-Squares estimator

Under these assumptions, the Ordinary-Least-Squares estimator for this model is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Once obtained, OLS analysis branches into two complementary aspects:

- Numerical: chapter 2.
- Statistical: chapter 3.

Chapter 2

Geometry of Linear Regression

2.1 The Geometry of Vector Spaces

Definition 2.1.1: Linear combination

A vector $\mathbf{z}$ is a linear combination of $\mathbf{x}_1$ and $\mathbf{x}_2$ if $\lambda_1\mathbf{x}_1 + \lambda_2\mathbf{x}_2 = \mathbf{z}, \quad \lambda_1, \lambda_2 \in \mathbb{R}$.

Example.

Let $\mathbf{x}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix}5\\6\\7\end{bmatrix}$, then $5\,\mathbf{x}_1 + \mathbf{x}_2 = \begin{bmatrix}10\\16\\22\end{bmatrix} = \mathbf{z}$.

Definition 2.1.2: Span

The set of all such combinations $\operatorname{span}\big(\mathbf{x}_1, \mathbf{x}_2\big) \;=\; S(\mathbf{x}_1, \mathbf{x}_2)$.

Definition 2.1.3: Standard basis of $\mathbb{R}^3$

$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ with $\mathbf{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$ spans the whole space. For any $\mathbf{z} \in \mathbb{R}^3$, $\mathbf{z} = \lambda_1\mathbf{e}_1 + \lambda_2\mathbf{e}_2 + \lambda_3\mathbf{e}_3, \qquad \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$

Example.

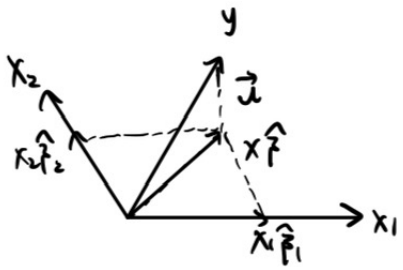
$$\mathbf{z} = \begin{bmatrix}10\\5\\-2\end{bmatrix}, \qquad \lambda_1 = 10,\ \lambda_2 = 5,\ \lambda_3 = -2$$

In the multiple-regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ with two regressors, where $\mathbf{X} = \begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2\end{bmatrix}$ $(m \times 2)$ and $\mathbf{y} \in \mathbb{R}^m$.
Since $\beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 = \mathbf{X}\boldsymbol{\beta}$, $\mathbf{X}\boldsymbol{\beta} \in S(\mathbf{X}) \subseteq \mathbb{R}^m$. By OLS (or MM),

$$\mathbf{x}_i^\top\big(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\big) = 0 \Rightarrow \mathbf{x}_i^\top\hat{\mathbf{u}} = 0 \Leftrightarrow \langle \mathbf{x}_i, \hat{\mathbf{u}}\rangle = 0 \Leftrightarrow \mathbf{x}_i \perp \hat{\mathbf{u}}$$

That is OLS gives us the minimized $\|\mathbf{u}\|$ and then $\hat{\mathbf{u}}$ is orthogonal to the $S(\mathbf{X})$ plane. Thus, by Pythagoras' Theorem (kindergarten knowledge),

$$\|\mathbf{y}\|^2 = \|\mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \|\hat{\mathbf{u}}\|^2 \Leftrightarrow \text{TSS} = \text{ESS} + \text{SSR}$$



Next, we will derive this equality formally.

2.2 The Geometry of OLS Estimation

2.2.1 Orthogonal Projections

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ with $m > K$. Then $\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}} = \hat{\mathbf{y}} + \hat{\mathbf{u}}$, where $\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\mathbf{y}}$ is the fitted value and $\hat{\mathbf{u}}$ is the residual. Since $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$,

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} = \mathbf{P}_\mathbf{X}\mathbf{y}, \text{ where we define } \mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$$

Definition 2.2.1: Projection (hat) matrix

$\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ is called projection matrix and it is $m \times m$ dimension.

Theorem 2.2.2

- $\mathbf{P}_\mathbf{X}\mathbf{X} = \mathbf{X}$. Intuitively, the projection of any matrix along its own hyperplane is itself.
- Symmetric: $\mathbf{P}_\mathbf{X}^\top = \mathbf{P}_\mathbf{X}$
- Idempotent $\mathbf{P}_\mathbf{X}^2 = \mathbf{P}_\mathbf{X}$

Proof.
- $\mathbf{P}_\mathbf{X}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{X}$
- $\mathbf{P}_\mathbf{X}^\top = \left[\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right]^\top = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{P}_\mathbf{X}$
- $\mathbf{P}_\mathbf{X}^2 = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{P}_\mathbf{X}$

□

$\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{I}\mathbf{y} - \mathbf{P}_\mathbf{X}\mathbf{y} = (\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{y} =: \mathbf{M}_\mathbf{X}\mathbf{y}$

Definition 2.2.3: Residual-maker matrix

$\mathbf{M}_\mathbf{X} = (\mathbf{I} - \mathbf{P}_\mathbf{X})$ is called residual-maker matrix.

Theorem 2.2.4

- $\mathbf{M}_\mathbf{X}\mathbf{X} = \mathbf{O}_{m\times K}$
- Symmetric: $\mathbf{M}_\mathbf{X}^\top = \mathbf{M}_\mathbf{X}$
- Orthogonal: $\hat{\mathbf{u}}^\top\mathbf{X} = 0$
- Idempotent: $\mathbf{M}_\mathbf{X}^2 = \mathbf{M}_\mathbf{X}$
- $\mathbf{M}_\mathbf{X}\hat{\mathbf{u}} = \hat{\mathbf{u}}$

Proof.

- $\mathbf{M_X X} = (\mathbf{I} - \mathbf{P_X})\mathbf{X} = \mathbf{X} - \mathbf{P_X X} = \mathbf{X} - \mathbf{X} = \mathbf{O}_{m\times m}$
- Both $\mathbf{I}$ and $\mathbf{P_X}$ are symmetric, then done.
- $\hat{\mathbf{u}}^\top \mathbf{X} = (\mathbf{M_X y})^\top \mathbf{X} = \mathbf{y}^\top \mathbf{M_X^\top X} = 0$
- $\mathbf{M_X^2} = (\mathbf{I} - \mathbf{P_X})(\mathbf{I} - \mathbf{P_X}) = \mathbf{I} - \mathbf{P_X} - \mathbf{P_X} + \mathbf{P_X^2} = \mathbf{I} - \mathbf{P_X} = \mathbf{M_X}$
- $\mathbf{M_X}\hat{\mathbf{u}} = \mathbf{M_X}(\mathbf{M_X y}) = \mathbf{M_X y} = \hat{\mathbf{u}}$

□

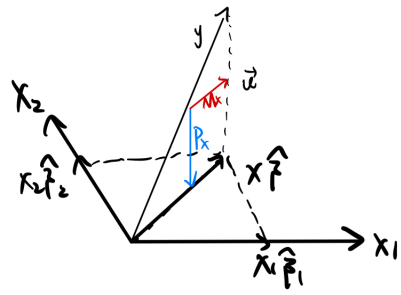


Figure 2.1: The effect of projection matrices.

Theorem 2.2.5: Complementary orthogonal projection

$\mathbf{y} = \mathbf{P_X y} + \mathbf{M_X y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$. That is the pair of projections $\mathbf{P_X}$ and $\mathbf{M_X}$ are said to be complementary projections. Any $\mathbf{y}$ can be decomposed into $\mathbf{P_X y}$ and $\mathbf{M_X y}$.

Proof. Since $\mathbf{M_X} = \mathbf{I} - \mathbf{P_X}$, $\mathbf{I} = \mathbf{P_X} + \mathbf{M_X}$. Then

$$\mathbf{y} = \mathbf{I y} = (\mathbf{P_X} + \mathbf{M_X})\,\mathbf{y} = \mathbf{P_X y} + \mathbf{M_X y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

□

Theorem 2.2.6

$\mathbf{P_X M_X} = \mathbf{0}$

Proof. $\mathbf{P_X M_X} = \mathbf{P_X}(\mathbf{I} - \mathbf{P_X}) = \mathbf{P_X} - \mathbf{P_X P_X} = \mathbf{P_X} - \mathbf{P_X} = \mathbf{0}$ □

Theorem 2.2.7

TSS=ESS+SSR.

Proof.

$$\|\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{y} = \langle \mathbf{y}, \mathbf{y} \rangle = \sum_{i=1}^{m} y_i^2 = \text{TSS}$$

and

$$\begin{aligned}
\text{TSS} = \mathbf{y}^\top \mathbf{y} &= (\mathbf{P_X y} + \mathbf{M_X y})^\top (\mathbf{P_X y} + \mathbf{M_X y}) \\
&= (\mathbf{y}^\top \mathbf{P_X} + \mathbf{y}^\top \mathbf{M_X})(\mathbf{P_X y} + \mathbf{M_X y}) \\
&= \mathbf{y}^\top \mathbf{P_X P_X y} + \mathbf{y}^\top \mathbf{P_X M_X y} + \mathbf{y}^\top \mathbf{M_X P_X y} + \mathbf{y}^\top \mathbf{M_X M_X y} \\
&= \mathbf{y}^\top \mathbf{P_X y} + \mathbf{y}^\top \mathbf{M_X y} \\
&= \|\mathbf{P_X y}\|^2 + \|\mathbf{M_X y}\|^2 \\
&= \|\mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \|\hat{\mathbf{u}}\|^2 = \text{ESS} + \text{SSR}.
\end{aligned}$$

where $\|\mathbf{P_X y}\|^2 = \mathbf{y}^\top \mathbf{P_X^\top P_X y} = \mathbf{y}^\top \mathbf{P_X P_X y} = \mathbf{y}^\top \mathbf{P_X y}$ and same for $\|\mathbf{M_X y}\|^2$. □

2.2.2 Linear Transformations of Regressors

Definition 2.2.8: Linear Transformation

To express any affine mapping $\mathbf{Y} = a\mathbf{C} + \mathbf{1}_m\mathbf{b}$ as a linear transformation: $\mathbf{X} \to \mathbf{XA}$

$$\mathbf{XA} = \begin{bmatrix}\mathbf{1}_m & \mathbf{C}\end{bmatrix}\begin{bmatrix}1 & b\\ 0 & a\end{bmatrix} = \begin{bmatrix}\mathbf{1}_m & a\mathbf{C} + \mathbf{1}_m\mathbf{b}\end{bmatrix} = \begin{bmatrix}\mathbf{1}_m & \mathbf{Y}\end{bmatrix}$$

Here $\mathbf{A}$ is full-rank then it is invertible and $\mathbf{A}^{-1}$ exists.
Original model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$. Choose any nonsingular $\mathbf{A} \in \mathbb{R}^{k\times k}$ and set the transformed regressor matrix $\mathbf{XA}$

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\big(\mathbf{A}\mathbf{A}^{-1}\big)\boldsymbol{\beta} = \mathbf{XA}\,\underbrace{\mathbf{A}^{-1}\boldsymbol{\beta}}_{\boldsymbol{\alpha}}$$

Then we get $\mathbf{y} = \mathbf{XA}\boldsymbol{\alpha} + \mathbf{u}$, where $\mathbf{A}^{-1}\boldsymbol{\beta} = \boldsymbol{\alpha}$. This means the coefficient of the regression changes.

Lemma 2.2.9

$\mathbf{P}_{\mathbf{XA}} = \mathbf{P}_{\mathbf{X}}$. That is $\mathbf{P}_{\mathbf{X}}$ is invariant to linear transformation.

Here $'$ is transpose.

Proof.

$$\mathbf{P}_{\mathbf{XA}} = \mathbf{XA}\,\underbrace{\big(\mathbf{A}'\mathbf{X}'\mathbf{XA}\big)^{-1}}_{k\times k}\mathbf{A}'\mathbf{X}' = \mathbf{XA}\mathbf{A}^{-1}\big(\mathbf{X}'\mathbf{X}\big)^{-1}\big(\mathbf{A}'\big)^{-1}\mathbf{A}'\mathbf{X}' = \mathbf{X}\big(\mathbf{X}'\mathbf{X}\big)^{-1}\mathbf{X}' = \mathbf{P}_{\mathbf{X}}$$

□

And $\mathbf{M}_{\mathbf{X}}$ is also invariant.

Theorem 2.2.10

Fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\hat{\mathbf{u}}$ are invariant to any nonsingular linear transformation. And then, $TSS = ESS + SSR$ is also invariant.

Proof. By lemma, it is clear that $\mathbf{XA}\hat{\boldsymbol{\alpha}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$. □

An important implication of the invariance of residuals and fitted values under nonsingular transformations of the regressors is that these vectors are unchanged if we change the units of measurement of the regressors.

Example.

> Let $\mathbf{F} = 32 + \frac{9}{5}\,\mathbf{C}$ and $\mathbf{X} = \begin{bmatrix}\mathbf{1}_m & \mathbf{C}\end{bmatrix}$.

Then

$$\mathbf{XA} = \underbrace{\begin{bmatrix}\mathbf{1}_m & \mathbf{C}\end{bmatrix}}_{m\times 2}\underbrace{\begin{bmatrix}1 & 32\\ 0 & \frac{9}{5}\end{bmatrix}}_{2\times 2} = \begin{bmatrix}\mathbf{1}_m + \mathbf{C}\cdot 0 & \mathbf{1}_m\cdot 32 + \frac{9}{5}\mathbf{C}\end{bmatrix} = \begin{bmatrix}\mathbf{1}_m & \mathbf{F}\end{bmatrix}$$

Then we transfer Celsius degrees into Fahrenheit degrees.

2.3 Frisch–Waugh–Lovell Theorem

2.3.1 Deviations from the Mean

Let's consider the model: $\mathbf{y} = \mathbf{1}_m \beta_1 + \mathbf{x}\beta_2 + \mathbf{u}$. In general, $\mathbf{1}_m \not\perp \mathbf{x}$ unless $\sum_t x_t = 0$. Sample mean is $\bar{x} = \frac{1}{m}\sum_t x_t$. Define the demeaned regressor $\mathbf{z} = \mathbf{x} - \bar{x}\,\mathbf{1}_m = \begin{bmatrix} x_1 - \bar{x} \\ \vdots \\ x_m - \bar{x} \end{bmatrix}$. Then we have $\mathbf{1}_m \perp \mathbf{z}$. Indeed,

$$\mathbf{z}'\mathbf{1}_m = (\mathbf{x} - \bar{x}\,\mathbf{1}_m)'\mathbf{1}_m = \mathbf{x}'\mathbf{1}_m - \bar{x}\,\mathbf{1}_m'\mathbf{1}_m = \sum_t x_t - \bar{x}m = 0$$

Because there exist an invertible $\mathbf{A} = \begin{bmatrix} 1 & -\bar{x} \\ 0 & 1 \end{bmatrix}$ such that $\begin{bmatrix} \mathbf{1}_m & \mathbf{x} \end{bmatrix}\mathbf{A} = \begin{bmatrix} \mathbf{1}_m & \mathbf{z} \end{bmatrix}$, we get a new model:

$$\mathbf{y} = \mathbf{1}_m\,\alpha_1 + \mathbf{z}\,\alpha_2 + \mathbf{u}$$

Definition 2.3.1

$\mathbf{P}_1 = \mathbf{1}_m\,(\mathbf{1}_m'\mathbf{1}_m)^{-1}\mathbf{1}_m'$

Then

$$\mathbf{P}_1 = \mathbf{1}_m\,(\mathbf{1}_m'\mathbf{1}_m)^{-1}\mathbf{1}_m' = \frac{1}{m}\,\mathbf{1}_m\mathbf{1}_m' = \frac{1}{m}\begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix}$$

Definition 2.3.2

$\mathbf{M}_1 = \mathbf{I} - \mathbf{P}_1$

Then

$$\mathbf{M}_1\mathbf{x} = (\mathbf{I} - \mathbf{P}_1)\mathbf{x} = \mathbf{I}\mathbf{x} - \mathbf{P}_1\mathbf{x} = \mathbf{x} - \frac{1}{m}\,\mathbf{1}_m\mathbf{1}_m'\mathbf{x} = \mathbf{x} - \frac{1}{m}\sum_t x_t\,\mathbf{1}_m = \mathbf{x} - \bar{x}\,\mathbf{1}_m = \mathbf{z}$$

Thus $\mathbf{M}_1$ is the matrix that converts any vector into its deviation-from-mean form. And we know $\mathbf{M}_1\mathbf{x} \perp \mathbf{1}_m$. The original model is

$$\mathbf{y} = \mathbf{1}_m\beta_1 + \mathbf{x}\beta_2 + \mathbf{u} \tag{2.1}$$

$$= \mathbf{1}_m\beta_1 + \beta_2\,\bar{x}\,\mathbf{1}_m - \beta_2\,\bar{x}\,\mathbf{1}_m + \mathbf{x}\beta_2 + \mathbf{u} = \mathbf{1}_m\big(\beta_1 + \beta_2\bar{x}\big) + \beta_2\,\big(\mathbf{x} - \bar{x}\,\mathbf{1}_m\big) + \mathbf{u} = \mathbf{1}_m\alpha_1 + \alpha_2\mathbf{z} + \mathbf{u},$$

where $\alpha_1 = \beta_1 + \beta_2\bar{x}$, $\alpha_2 = \beta_2$, and $\mathbf{z} = \mathbf{x} - \bar{x}\,\mathbf{1}_m$. Since $\mathbf{z} = \mathbf{M}_1\mathbf{x}$,

$$\mathbf{y} = \mathbf{1}_m\alpha_1 + \beta_2\mathbf{M}_1\mathbf{x} + \mathbf{u}$$

Then we define $\mathbf{v} = \mathbf{u} + \mathbf{1}_m\alpha_1$,

$$\mathbf{y} = \mathbf{z}\beta_2 + \mathbf{v} = \mathbf{M}_1\mathbf{x}\beta_2 + \mathbf{v} \tag{2.2}$$

If we times $\mathbf{M}_1$ on the left of (2.1),

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1(\mathbf{1}_m\beta_1 + \mathbf{x}\beta_2 + \mathbf{u}) = \mathbf{M}_1\mathbf{x}\beta_2 + \mathbf{M}_1\mathbf{u} \tag{2.3}$$

Do OLS for (2.1) and we get:

$$\boldsymbol{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = (\mathbf{x}^\top\mathbf{x})^{-1}\mathbf{x}^\top\mathbf{y}$$

Do OLS for (2.3) and we get:

$$\hat{\beta}_2 = \frac{\mathbf{x}^\top\mathbf{M}_1\mathbf{y}}{\mathbf{x}^\top\mathbf{M}_1\mathbf{x}} = \frac{\mathbf{x}^\top\mathbf{M}_1^\top\mathbf{M}_1\mathbf{y}}{\mathbf{x}^\top\mathbf{M}_1^\top\mathbf{M}_1\mathbf{x}} = \frac{\sum_t(x_t - \bar{x})(y_t - \bar{y})}{\sum_t(x_t - \bar{x})^2} = \frac{\mathrm{Cov}(\mathbf{x}, \mathbf{y})}{\mathrm{Var}(\mathbf{x})}$$

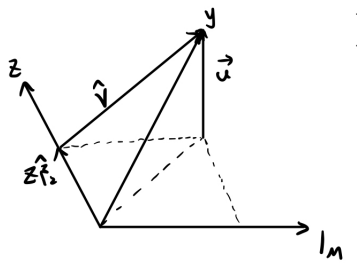


Figure 2.2: Geometry interpretation of equation (2.2)

2.3.2 Two Groups of Regressors

Consider the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}$, where $\mathbf{X}_1 \in \mathbb{R}^{m\times k_1}$ and $\mathbf{X}_2 \in \mathbb{R}^{m\times k_2}$ and $k_1 + k_2 = k, k_1,\ k_2 \geq 0$. We assume the orthogonality assumption (we will relax it later): $\mathbf{X}_1 \perp \mathbf{X}_2 \Rightarrow \mathbf{X}_2'\mathbf{X}_1 = 0$. There are three Projection (hat) matrices:

- $\mathbf{P_X} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$
- $\mathbf{y} = \mathbf{X}_1\boldsymbol{\alpha}_1 + \mathbf{v}_1 \quad \Rightarrow \quad \mathbf{P}_1 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$
- $\mathbf{y} = \mathbf{X}_2\boldsymbol{\alpha}_2 + \mathbf{v}_2 \quad \Rightarrow \quad \mathbf{P}_2 = \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$

Theorem 2.3.3

- $\mathbf{P_X}\mathbf{P}_1 = \mathbf{P}_1 = \mathbf{P}_1\mathbf{P_X}$. Same for $\mathbf{P_X}\mathbf{P}_2$.
- $\mathbf{M}_1\mathbf{X}_1 = 0$
- $\mathbf{M}_1\mathbf{M_X} = \mathbf{M_X}$
- Under $\mathbf{X}_1 \perp \mathbf{X}_2$, $\mathbf{P}_1\mathbf{X}_2 = 0$ and $\mathbf{M}_1\mathbf{X}_2 = \mathbf{X}_2$

Proof.

- $\mathbf{P_X}\mathbf{P}_1 = \mathbf{P_X}\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1' = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1' = \mathbf{P}_1 = \mathbf{P}_1\mathbf{P_X}$
- $\mathbf{M}_1\mathbf{X}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{P}_1\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{X}_1 = 0$
- $\mathbf{M}_1\mathbf{M_X} = (\mathbf{I} - \mathbf{P}_1)(\mathbf{I} - \mathbf{P_X}) = \mathbf{I} - \mathbf{P}_1 - \mathbf{P_X} + \mathbf{P}_1\mathbf{P_X} = \mathbf{I} - \mathbf{P_X} = \mathbf{M_X}$
- $\mathbf{P}_1\mathbf{X}_2 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2 = 0$. $\mathbf{M}_1\mathbf{X}_2 = (\mathbf{I}_1 - \mathbf{P}_1)\mathbf{X}_2 = \mathbf{X}_2$

□

Theorem 2.3.4

Under $\mathbf{X}_1 \perp \mathbf{X}_2$, we have $\hat{\alpha_1} = \hat{\beta}_1$ and $\hat{\alpha_2} = \hat{\beta}_2$.

Proof.

$$\hat{\mathbf{y}} = \mathbf{X}_1\hat{\alpha_1} = \mathbf{P}_1\mathbf{y} = \mathbf{P}_1\mathbf{P_X}\mathbf{y} = \mathbf{P}_1\big(\mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2\big) = \mathbf{P}_1\mathbf{X}_1\hat{\beta}_1 + \mathbf{P}_1\mathbf{X}_2\hat{\beta}_2 = \mathbf{X}_1\hat{\beta}_1 + \mathbf{P}_1\mathbf{X}_2\hat{\beta}_2 = \mathbf{X}_1\hat{\beta}_1$$

$$\Rightarrow \mathbf{X}_1\hat{\alpha_1} = \mathbf{X}_1\hat{\beta}_1$$

Similar for $\hat{\alpha_2} = \hat{\beta}_2$. □

Thus, under $\mathbf{X}_1 \perp \mathbf{X}_2$, we have

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u} \quad \longrightarrow \quad \hat{\boldsymbol{\beta}} = \big(\mathbf{X}'\mathbf{X}\big)^{-1}\mathbf{X}'\mathbf{y} = \begin{bmatrix}\hat{\beta}_1\\ \hat{\beta}_2\end{bmatrix}$$

$$\hat{\mathbf{y}} = \mathbf{X}_1\boldsymbol{\alpha}_1 + \mathbf{v}_1 \quad \longrightarrow \quad \hat{\alpha}_1 = \big(\mathbf{X}_1'\mathbf{X}_1\big)^{-1}\mathbf{X}_1'\mathbf{y} = \hat{\beta}_1$$

$$\hat{\mathbf{y}} \;=\; \mathbf{X}_2\boldsymbol{\alpha}_2 \;+\; \mathbf{v}_2 \quad\longrightarrow\quad \hat{\alpha}_2 \;=\; \left(\mathbf{X}_2'\mathbf{X}_2\right)^{-1}\mathbf{X}_2'\mathbf{y} \;=\; \hat{\beta}_2$$

Now consider model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}$ with $\mathbf{X}_1 \not\perp \mathbf{X}_2$:

Lemma 2.3.5

With the transformation $\mathbf{y} = \mathbf{X}_1\alpha_1 + (\mathbf{X}_2 + \mathbf{X}_1\mathbf{A})\alpha_2 + \mathbf{u}$, we have $\hat{\alpha}_2 = \hat{\boldsymbol{\beta}}_2$.

Proof.

$$\mathbf{y} = \mathbf{X}_1\alpha_1 + (\mathbf{X}_2 + \mathbf{X}_1\mathbf{A})\alpha_2 + \mathbf{u} = \mathbf{X}_1\alpha_1 + \mathbf{X}_2\alpha_2 + \mathbf{X}_1\mathbf{A}\alpha_2 + \mathbf{u} = \mathbf{X}_1[\,\alpha_1 + \mathbf{A}\alpha_2\,] + \mathbf{X}_2\alpha_2 + \mathbf{u}$$

Clearly, $\hat{\alpha}_2 = \hat{\boldsymbol{\beta}}_2$. □

Recall. $\mathbf{M}_1 = \mathbf{I} - \mathbf{P}_1 = \mathbf{I} - \mathbf{X}_1\left(\mathbf{X}_1^\intercal\mathbf{X}_1\right)^{-1}\mathbf{X}_1^\intercal$. It is clear that $\mathbf{M}_1\mathbf{X}_2 \perp \mathbf{X}_1$. Consider the model:

$$\mathbf{y} = \mathbf{X}_1\alpha_1 + \mathbf{M}_1\mathbf{X}_2\alpha_2 + \mathbf{u} = \mathbf{X}_1\alpha_1 + \left[\mathbf{X}_2 - \mathbf{X}_1\left(\mathbf{X}_1^\intercal\mathbf{X}_1\right)^{-1}\mathbf{X}_1^\intercal\mathbf{X}_2\right]\alpha_2 + \mathbf{u}$$

By lemma above, $\alpha_2 = \boldsymbol{\beta}_2$. So,

$$\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{v}, \text{ where } \mathbf{v} = \mathbf{u} + \mathbf{X}_1\alpha_1. \tag{2.4}$$

If we times $\mathbf{M}_1$ on the left of (2.4):

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{M}_1\left(\mathbf{u} + \mathbf{X}_1\alpha_1\right) = \mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{M_1}\mathbf{u}.$$

To see the difference:

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u} \quad\rightarrow\quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{y} = \begin{bmatrix}\hat{\boldsymbol{\beta}}_1\\ \hat{\boldsymbol{\beta}}_2\end{bmatrix}$$

Using the above transformation

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{M_1}\mathbf{u} \quad\rightarrow\quad \hat{\boldsymbol{\beta}}_2 = \left(\mathbf{X}_2^\intercal\mathbf{M}_1^\intercal\mathbf{M}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2^\intercal\mathbf{M}_1^\intercal\mathbf{M}_1\mathbf{y} = \left(\mathbf{X}_2^\intercal\mathbf{M}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2^\intercal\mathbf{M}_1\mathbf{y}$$

However,

$$\mathbf{y} = \mathbf{X}_2\boldsymbol{\alpha}_2 + \mathbf{v} \quad\rightarrow\quad \boldsymbol{\alpha}_2 = (\mathbf{X}_2^\intercal\mathbf{X}_2)^{-1}\mathbf{X}_2^\intercal\mathbf{y} \neq \hat{\boldsymbol{\beta}}_2$$

Similar for α_2 and β_2:

$$\mathbf{M}_2\mathbf{y} = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\alpha}_1 + \mathbf{M_1}\mathbf{u} = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{M_1}\mathbf{u} \quad\rightarrow\quad \hat{\boldsymbol{\beta}}_1 = \left(\mathbf{X}_1^\intercal\mathbf{M}_2\mathbf{X}_1\right)^{-1}\mathbf{X}_1^\intercal\mathbf{M}_2\mathbf{y}$$

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\alpha}_1 + \mathbf{v} \quad\rightarrow\quad \hat{\boldsymbol{\alpha}}_1 = (\mathbf{X}_1^\intercal\mathbf{X}_1)^{-1}\mathbf{X}_1^\intercal\mathbf{y} \neq \hat{\boldsymbol{\beta}}_1$$

Formally, we have the following theorem.

Theorem 2.3.6: Frisch-Waugh-Lovell Theorem

With model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}$, premultiply everything by $\mathbf{M}_1 = \mathbf{I} - \mathbf{P}_1$ and then

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\alpha_2 + \mathbf{v}$$

We have

(i) $\hat{\beta}_2 = \hat{\alpha}_2$

(ii) $\hat{\mathbf{u}} = \hat{\mathbf{v}}$

Proof. (i)

$$\hat{\alpha}_2 \;=\; \left((\mathbf{M}_1\mathbf{X}_2)'\mathbf{M}_1\mathbf{X}_2\right)^{-1}(\mathbf{M}_1\mathbf{X}_2)'\mathbf{M}_1\mathbf{y} \;=\; \left(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y}$$

because $\mathbf{M}_1' = \mathbf{M}_1, \mathbf{M}_1\mathbf{M}_1 = \mathbf{M}_1$.

$$\mathbf{y} = \mathbf{P_X y} + \mathbf{M_X y} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2 + \mathbf{M_X y}$$

Then

$$\mathbf{X}_2'\mathbf{M}_1\mathbf{y} = \mathbf{X}_2'\mathbf{M}_1\mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 + \mathbf{X}_2'\mathbf{M}_1\mathbf{M_X y}$$

We claim $\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_1 = 0$ and $\mathbf{X}_2'\mathbf{M}_1\mathbf{M_X} = 0$. Indeed,

$$\mathbf{M}_1\mathbf{X}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{P}_1\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_1 = \mathbf{X}_1 - \mathbf{X}_1 = 0$$

and

$$\mathbf{M}_1\mathbf{M_X} = (\mathbf{I} - \mathbf{P}_1)(\mathbf{I} - \mathbf{P_X}) = \mathbf{I} - \mathbf{P}_1 - \mathbf{P_X} + \mathbf{P}_1\mathbf{P_X} = \mathbf{I} - \mathbf{P_X} = \mathbf{M_X}$$

Thus,

$$\mathbf{X}_2'\mathbf{M}_1\mathbf{y} = \mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 \quad \Longrightarrow \quad \hat{\beta}_2 = \left(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y} = \hat{\alpha}_2$$

(ii) Since $\mathbf{y} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2 + \hat{\mathbf{u}}$ and $\hat{\mathbf{u}} = \mathbf{M_X y}$,

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_1\hat{\beta}_1 + \mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 + \mathbf{M}_1\mathbf{M_X y} = 0 + \mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 + \mathbf{M_X y} = \mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 + \hat{\mathbf{u}}$$

Thus, $\hat{\mathbf{u}} = \hat{\mathbf{v}}$.

□

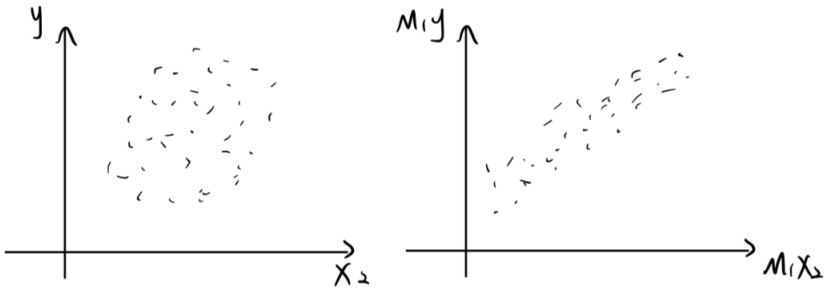


Figure 2.3: Graphical explanation: with FWL, we can create an orthogonal environment to estimate β_2.

2.4 Applications of the FWL Theorem

2.4.1 Goodness of Fit

Recall. From $\mathbf{y} = \hat{\mathbf{y}} + \hat{\mathbf{u}}$ and $\hat{\mathbf{y}} \perp \hat{\mathbf{u}}$, we know

$$\|\mathbf{y}\|^2 = \|\mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \|\hat{\mathbf{u}}\|^2 \quad \Leftrightarrow \quad \text{TSS} = \text{ESS} + \text{SSR}$$

where $\|\mathbf{y}\|^2 = \sum_t y_t^2$ and $\|\hat{\mathbf{y}}\|^2 = \sum_t \hat{y}_t^2$.

Definition 2.4.1

$R_u^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}, \quad 0 \le R_u^2 \le 1$. This is the uncentered coefficient of determination.

Add a vertical shift $\mathbf{1}_m\alpha$. Then the model becomes

$$\mathbf{y} - \mathbf{1}_m\alpha = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad \Leftrightarrow \quad \mathbf{y} = \mathbf{1}_m\alpha + \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad (\alpha = 0 \;\Rightarrow\; \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}).$$

Then

$$\mathbf{y} - \mathbf{1}_m\alpha = \mathbf{P_X}\left(\mathbf{y} - \mathbf{1}_m\alpha\right) + \mathbf{M_X}\left(\mathbf{y} - \mathbf{1}_m\alpha\right) = \mathbf{P_X y} - \mathbf{P_X}\mathbf{1}_m\alpha + \mathbf{M_X y} - \mathbf{M_X}\mathbf{1}_m\alpha$$

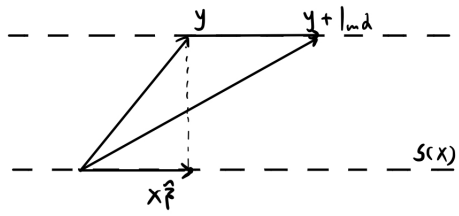
Because $\mathbf{P_X 1}_m = \mathbf{1}_m$, $\mathbf{M_X 1}_m = \mathbf{0}$,

$$\mathbf{y} - \mathbf{1}_m\alpha = \mathbf{P_X y} - \mathbf{1}_m\alpha + \mathbf{M_X y}$$

Hence the uncentred coefficient of determination is

$$R_u^2 = \frac{\left\|\mathbf{P_X y} - \mathbf{1}_m\alpha\right\|^2}{\left\|\mathbf{y} - \mathbf{1}_m\alpha\right\|^2}$$

By choosing α sufficiently large, we can in fact make R_u^2 as close as we wish to 1, because, for very large α, the term $\alpha\mathbf{1}_m$ will completely dominate the terms $\mathbf{P_X y}$ and $\mathbf{y}$ in the numerator and denominator respectively. But a large R_u^2 in such a case would be entirely misleading, since the "good fit" would be accounted for almost exclusively by the constant.



This can be solved by FWL. Since $\mathbf{M}_1 = \mathbf{I} - \mathbf{P}_1 = \mathbf{I} - \mathbf{1}_m\left(\mathbf{1}_m'\mathbf{1}_m\right)^{-1}\mathbf{1}_m'$,

$$\mathbf{M}_1\left(\mathbf{y} - \mathbf{1}_m\alpha\right) = \mathbf{M}_1\mathbf{y} - \alpha\mathbf{M}_1\mathbf{1}_m = \mathbf{M}_1\mathbf{y}$$

Hence the centred coefficient of determination is

$$R_c^2 = \frac{\left\|\mathbf{P_X M}_1\mathbf{y}\right\|^2}{\left\|\mathbf{M}_1\mathbf{y}\right\|^2}$$

Theorem 2.4.2

R_c^2 is invariant to linear transformations of $\mathbf{X}$ and $\mathbf{y}$.

2.4.2 Influential Observations and Leverage

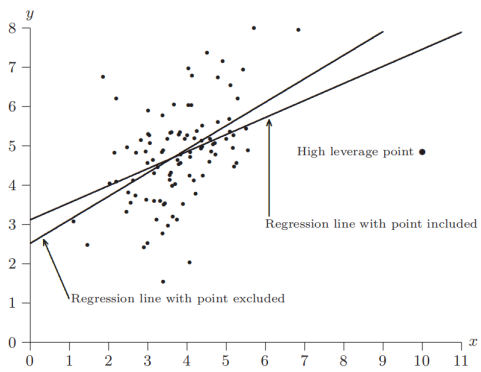


Figure 2.4: An influential observation

The regression lines for the two regressions in Figure 2.4 are substantially different. The high leverage point is quite distant from the regression line obtained when it is excluded. When that point is included, it is able, by virtue of its position well to the right of the other observations, to exert a good deal of **leverage** on the regression line, pulling it down toward itself. **It is the x coordinate that gives the point its position of high leverage, but it is the y coordinate that determines whether the**

high leverage position will actually be exploited, resulting in substantial influence on the regression line.
We could use trimming or quantile regression to address this.

Leverage

Now we want to estimate the effect of a single observation on $\boldsymbol{\beta}$. It turns out we can use FWL to estimate that.

Definition 2.4.3

Indicator vector for observation t is $\mathbf{e}_t = (0, \dots, 1, \dots, 0)^\top_{m\times 1}$.

Clearly, $\mathbf{e}_t'\mathbf{e}_t = 1$.

$$\begin{cases} \text{Regression line with point excluded: } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \\ \text{Regression line with point included: } \mathbf{y} = \mathbf{X}\boldsymbol{\beta}^{(t)} + \alpha\,\mathbf{e}_t + \mathbf{v} \end{cases}$$

By FWL, $\mathbf{M}_t\mathbf{y} = \mathbf{M}_t\mathbf{X}\boldsymbol{\beta}^{(t)} + \mathbf{v}$, where $\mathbf{M}_t = \mathbf{I} - \mathbf{P}_t = \mathbf{I} - \mathbf{e}_t\big(\mathbf{e}_t'\mathbf{e}_t\big)^{-1}\mathbf{e}_t' = \mathbf{I} - \mathbf{e}_t\mathbf{e}_t'$. Thus,

$$\mathbf{M}_t\mathbf{y} = \big(\mathbf{I} - \mathbf{e}_t\mathbf{e}_t'\big)\mathbf{y} = \mathbf{y} - \mathbf{e}_t\,\mathbf{e}_t'\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_t \\ \vdots \\ y_m \end{bmatrix} - \begin{bmatrix} 0 \\ \vdots \\ y_t \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ 0 \\ \vdots \\ y_m \end{bmatrix}$$

We want to know the difference between $\boldsymbol{\beta}^{(t)}$ and $\boldsymbol{\beta}$. Combine two models together

$$\begin{cases} \mathbf{y} \;=\; \mathbf{X}\boldsymbol{\beta}^{(t)} + \alpha\,\mathbf{e}_t + \mathbf{v}, \\ \mathbf{y} \;=\; \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \end{cases} \qquad \mathbf{Z} = \begin{bmatrix} \mathbf{X} & \mathbf{e}_t \end{bmatrix}$$

Projection decomposition

$$\mathbf{y} \;=\; \mathbf{P}_{\mathbf{Z}}\mathbf{y} + \mathbf{M}_{\mathbf{Z}}\mathbf{y} \;=\; \mathbf{X}\hat{\boldsymbol{\beta}}^{(t)} + \hat{\alpha}\,\mathbf{e}_t + \mathbf{M}_{\mathbf{Z}}\mathbf{y}$$

Left-multiply by the hat matrix

$$\mathbf{P}_{\mathbf{X}}\mathbf{y} = \mathbf{P}_{\mathbf{X}}\mathbf{X}\hat{\boldsymbol{\beta}}^{(t)} + \hat{\alpha}\,\mathbf{P}_{\mathbf{X}}\mathbf{e}_t + \underbrace{\mathbf{P}_{\mathbf{X}}\mathbf{M}_{\mathbf{Z}}\mathbf{y}}_{=0} = \mathbf{X}\hat{\boldsymbol{\beta}}^{(t)} + \hat{\alpha}\,\mathbf{P}_{\mathbf{X}}\mathbf{e}_t$$

Thus,

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{P}_{\mathbf{X}}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}^{(t)} + \hat{\alpha}\,\mathbf{P}_{\mathbf{X}}\mathbf{e}_t \;\Leftrightarrow\; \mathbf{X}\hat{\boldsymbol{\beta}}^{(t)} - \mathbf{X}\hat{\boldsymbol{\beta}} = -\,\hat{\alpha}\,\mathbf{P}_{\mathbf{X}}\mathbf{e}_t \;\Leftrightarrow\; \mathbf{X}\big(\hat{\boldsymbol{\beta}}^{(t)} - \hat{\boldsymbol{\beta}}\big) = -\,\hat{\alpha}\,\mathbf{P}_{\mathbf{X}}\mathbf{e}_t$$

Using FWL,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^{(t)} + \alpha\,\mathbf{e}_t + \mathbf{v} \qquad \Longrightarrow \qquad \mathbf{M}_{\mathbf{X}}\mathbf{y} = \alpha\,\mathbf{M}_{\mathbf{X}}\mathbf{e}_t + \mathbf{v}$$

Then

$$\hat{\alpha} = \big(\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}'\mathbf{M}_{\mathbf{X}}\mathbf{e}_t\big)^{-1}\,\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}'\mathbf{M}_{\mathbf{X}}\mathbf{y} = \big(\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}\mathbf{e}_t\big)^{-1}\,\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}\mathbf{y} = \frac{\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}\mathbf{y}}{\mathbf{e}_t'\mathbf{M}_{\mathbf{X}}\mathbf{e}_t}$$

Definition 2.4.4

The tth diagonal element of the hat matrix $\mathbf{P}_{\mathbf{X}}$ is

$$H_t = \mathbf{e}_t'\mathbf{P}_{\mathbf{X}}\mathbf{e}_t = \mathbf{e}_t'\mathbf{P}_{\mathbf{X}}'\mathbf{P}_{\mathbf{X}}\mathbf{e}_t = \langle \mathbf{P}_{\mathbf{X}}\mathbf{e}_t,\ \mathbf{P}_{\mathbf{X}}\mathbf{e}_t\rangle = \|\mathbf{P}_{\mathbf{X}}\mathbf{e}_t\|^2$$

Theorem 2.4.5

$0 \leq H_t \leq 1$.

Proof. Since $\|\mathbf{P_X y}\|^2 \leq \|\mathbf{y}\|^2$, $\|\mathbf{P_X e}_t\|^2 \leq \|\mathbf{e}_t\|^2 = 1 \Longrightarrow 0 \leq H_t \leq 1$. □

Since $\mathbf{e}_t'\mathbf{M_X y} = \mathbf{e}_t'\hat{\mathbf{u}} = \hat{u}_t$ and $\mathbf{e}_t'\mathbf{M_X e}_t = 1 - H_t$, then we have

$$\hat{\alpha} = \frac{\hat{u}_t}{1 - H_t}$$

Back to the formula,

$$\mathbf{X}\big(\hat{\boldsymbol{\beta}}^{(t)} - \hat{\boldsymbol{\beta}}\big) = -\frac{\hat{u}_t}{1 - H_t}\mathbf{P_X}\,\mathbf{e}_t$$

times $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ on the left,

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbf{X}\big(\hat{\boldsymbol{\beta}}^{(t)} - \hat{\boldsymbol{\beta}}\big) = -\frac{\hat{u}_t}{1 - H_t}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{P_X e}_t$$

We get the leverage of tth observation,

$$\hat{\boldsymbol{\beta}}^{(t)} - \hat{\boldsymbol{\beta}} = -\frac{\hat{u}_t}{1 - H_t}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}_t$$

It is evident that the influence of an observation depends on both $\hat{u}_t$ and H_t. It will be greater if the observation has a large residual, which, as we saw in Figure 2.4, is related to its y coordinate. On the other hand, H_t is related to the x coordinate of a point, which, as we also saw in the figure, determines the leverage, or potential influence, of the corresponding observation. We say that observations for which H_t is large have high leverage or are leverage points. A leverage point is not necessarily influential, but it has the potential to be influential.

Theorem 2.4.6

$\sum_t H_t = K$, where K is the number of predictors including the intercept.

Proof. $\sum_t H_t = \text{tr}(\mathbf{P_X}) = \text{tr}\big(\mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\big)$. Since $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$, $\sum_t H_t = \text{tr}\big(\mathbf{X}'\mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\big) = \text{tr}(\mathbf{I}_K) = K$. □

Hence the average of all the H_t is $\frac{\sum_t H_t}{m} = \frac{K}{m}$. In a balanced design, $\forall t,\ H_t \approx \frac{K}{m}$.

Chapter 3

Statistical properties of OLS

3.1 Unbiasedness

Definition 3.1.1: Unbiasedness

$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} \;\Leftrightarrow\; E(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = 0$

Theorem 3.1.2

$\bar{X} = \frac{1}{m}\sum_t x_t$ is unbiased.

Proof.

$$E(\bar{X}) = E\left(\frac{1}{m}\sum_t x_t\right) = \frac{1}{m}\sum_t E(x_t) = \frac{1}{m}\sum_t \mu = \frac{1}{m} m\mu = \mu$$

□

Estimator decomposition

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\,\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}$$

Then

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}$$

Take expectation

$$E(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\big]$$

3.1.1 Assumptions about Error Terms and Regressors

Conditions that drive the expectation to zero

(a) $\mathbf{X}$ are fixed numbers. Indeed,

$$E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\big] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E\big[\mathbf{u}\big] = 0, \text{ since } E(\mathbf{u}) = 0.$$

Unfortunately, this is frequently not a reasonable assumption to make in applied econometric work.

(b) $\mathbf{X}$ are exogenous :

$$E(\mathbf{u} \mid \mathbf{X}) = 0 \tag{3.1}$$

Indeed,

$$E(\mathbf{u} \mid \mathbf{X}) = 0 \Rightarrow E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\big] = E_{\mathbf{X}}\Big\{\, E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \mid \mathbf{X}\big]\Big\} = E_{\mathbf{X}}\Big\{\, (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u} \mid \mathbf{X})\Big\} = 0$$

Assumption (3.1) is perfectly reasonable in the context of some types of data, like cross-section data. On the other hand, suppose that a sample consists of time-series data, u_t must be related to future values if current values of the dependent variable affect future values of some of the regressors. Thus, in the context of time-series data, the exogeneity assumption (1) is a very strong one that we may often not feel comfortable in making. Instead, we make a new assumption:

$$E(u_t \mid \mathbf{x}_t) = 0 \tag{3.2}$$

which is substantially weaker than assumption (1) because it merely rules out the possibility that it may depend on their values for the current observation.
It turns out that (3.1) implies (3.2) but not the other way around. Under assumption (3.2), OLS is usually biased but is consistent (we will see it later).

3.1.2 The OLS Estimator Can Be Biased

Consider an autoregressive model, namely AR(1),

$$\mathbf{y} = \beta_1 \mathbf{1} + \beta_2 \mathbf{y}_1 + \mathbf{u}, \quad \text{where } \mathbf{y} = \begin{bmatrix} y_2 \\ \vdots \\ y_M \end{bmatrix}, \quad \mathbf{X} = [\mathbf{1}, \mathbf{y}_1] = \begin{bmatrix} 1 & y_1 \\ \vdots & \vdots \\ 1 & y_{M-1} \end{bmatrix}.$$

In terms of a typical observation: $y_t = \beta_1 + \beta_2 y_{t-1} + u_t$.
It is perfectly reasonable to assume that assumption (3.2) holds, because this condition amounts to saying that $\mathrm{E}(u_t) = 0$ for every possible value of y_{t-1}. Not only is y_{t-1} realized before u_t, but its realized value has no impact on the expectation of u_t.
However, it is clear that the exogeneity assumption (3.1), which would here require that $\mathrm{E}(\mathbf{u} \mid \mathbf{y}_1) = 0$, cannot possibly hold, because y_{t-1} depends on u_{t-1}, u_{t-2}, and so on. Thus, assumption (3.1) will evidently fail to hold! To see the consequences of assumption (3.1) not holding:
By FWL, $\mathbf{P}_1 = \mathbf{1}_m(\mathbf{1}_m'\mathbf{1}_m)^{-1}\mathbf{1}_m'$ and $\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{y}_1\,\beta_2 + \mathbf{M}_1\mathbf{u}$. Thus,

$$\hat{\beta}_2 = (\mathbf{y}_1'\mathbf{M}_1\mathbf{y}_1)^{-1}\mathbf{y}_1'\mathbf{M}_1\mathbf{y}_1 = \beta_2 + (\mathbf{y}_1'\mathbf{M}_1\mathbf{y}_1)^{-1}\,\mathbf{y}_1'\mathbf{M}_1\mathbf{u}$$

It is clear that $\hat{\beta}_2$ will be unbiased if and only if the second term above has expectation zero. But this term does not have expectation zero.

3.2 Consistency

Definition 3.2.1

An estimator $\hat{\theta}_m$ is consistent if $\operatorname{plim}_{m\to\infty} \hat{\theta}_m = \theta$ ($\Leftrightarrow$ $\hat{\theta}_m \xrightarrow{p} \theta$). Formally, for $\varepsilon > 0$, $\lim_{m\to\infty} P\big(|\hat{\theta}_m - \theta| < \varepsilon\big) = 1$.

Theorem 3.2.2: Law of Large Numbers

If x_t with mean $E(x_t) = \mu$ and finite variance $\mathrm{Var}(x_t) = \sigma^2$, then the sample average $\bar{X} = \frac{1}{m}\sum_t x_t$ satisfies $\operatorname{plim}_{m\to\infty} \bar{X} = \mu$. That is, $\bar{X} = \frac{1}{m}\sum_t x_t$ is a consistent estimator for the population mean.

Definition 3.2.3: MSE convergence and consistency

Mean–squared error decomposes into squared bias plus variance:

$$\mathrm{MSE} = E\big((\hat{\theta}_m - \theta)^2\big) = \mathrm{Bias}^2 + \mathrm{Variance}$$

MSE converges iff MSE $\to 0$

Clearly, MSE convergence implies consistency.

Theorem 3.2.4

$\bar{X} = \frac{1}{m}\sum_t x_t$ is consistent.

Proof. Since $\mathrm{Cor}(x_t, x_s) = 0$,

$$\mathrm{Var}\left(\frac{1}{m}\sum_t x_t\right) = \frac{1}{m^2}\mathrm{Var}\left(\sum_t x_t\right) = \frac{1}{m^2}\sum_t \mathrm{Var}(x_t) = \frac{1}{m^2}\sum_t \sigma^2 = \frac{1}{m^2}m\sigma^2 = \frac{\sigma^2}{m} \to 0$$

Thus,

$$\mathrm{MSE}(\bar{X}) = 0 + \frac{\sigma^2}{m} \to 0$$

□

However, working with plims can be a little bit tricky. Consider the matrix $\mathbf{X}^\top\mathbf{X}$. As $m \to \infty$, we would expect that, in most circumstances, such a sum would tend to infinity as well. Therefore, the matrix $\mathbf{X}^\top\mathbf{X}$ will generally not have a plim. However, it is not at all unreasonable to assume that

$$\mathrm{plim}_{m\to\infty}\frac{1}{m}\mathbf{X}^\top\mathbf{X} = \mathbf{S}_{\mathbf{X}^\top\mathbf{X}} \tag{3.3}$$

where $\mathbf{S}_{\mathbf{X}^\top\mathbf{X}}$ is a nonstochastic matrix with full rank k, since each element of the matrix on the left-hand side is now an average of m numbers:

$$\left(\frac{1}{m}\mathbf{X}^\top\mathbf{X}\right)_{ij} = \frac{1}{m}\sum_{t=1}^{m} X_{ti}X_{tj}.$$

In effect, when we write (3.3), we are implicitly making some assumption sufficient for a LLN to hold for the sequences generated by the squares of the regressors and their cross-products.

Theorem 3.2.5

Under the assumptions $E(u_t \mid \mathbf{X}_t) = 0$ and $\mathrm{plim}_{m\to\infty}\frac{1}{m}\mathbf{X}^\top\mathbf{X} = \mathbf{S}_{\mathbf{X}^\top\mathbf{X}}$, the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is consistent.

Proof.

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \;\Rightarrow\; \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \left(\frac{\mathbf{X}'\mathbf{X}}{m}\right)^{-1}\frac{\mathbf{X}'\mathbf{u}}{m}$$

By assumption (3.3),

$$\mathrm{plim}_{m\to\infty}\frac{1}{m}\mathbf{X}^\top\mathbf{X} = \mathbf{S}_{\mathbf{X}^\top\mathbf{X}}.$$

Then we only need to show $\mathrm{plim}\,\frac{\mathbf{X}'\mathbf{u}}{m} = 0$. By assumption (3.2) $E(u_t \mid \mathbf{x}_t) = 0 \Rightarrow E(\mathbf{x}_t' u_t) = 0$,

$$\mathrm{plim}\,\frac{\mathbf{X}'\mathbf{u}}{m} = \mathrm{plim}\,\frac{1}{m}\sum_t \mathbf{x}_t' u_t = E(\mathbf{x}_t' u_t) = 0$$

Thus,

$$\mathrm{plim}\,\frac{m}{m}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} = \mathrm{plim}\left(\frac{\mathbf{X}'\mathbf{X}}{m}\right)^{-1}\frac{\mathbf{X}'\mathbf{u}}{m} = 0 \;\Rightarrow\; \mathrm{plim}\,\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}$$

□

3.3 The Covariance Matrix of the OLS Parameter Estimates

Definition 3.3.1: Variance

For a scalar random variable z:

$$\mathrm{Var}(z) = E\big(z - E(z)\big)^2$$

Definition 3.3.2: Covariance matrix

For a $k \times 1$ random vector $\mathbf{z}$:

$$\begin{bmatrix} \mathrm{Var}(z_1) & \mathrm{Cov}(z_1, z_2) & \cdots & \mathrm{Cov}(z_1, z_k) \\ \mathrm{Cov}(z_2, z_1) & \mathrm{Var}(z_2) & \cdots & \mathrm{Cov}(z_2, z_k) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Var}(z_k) & \mathrm{Cov}(z_k, z_2) & \cdots & \mathrm{Var}(z_k) \end{bmatrix}$$

a symmetric matrix. This matrix is called the covariance matrix, although the terms variance matrix and variance-covariance matrix are also sometimes used.

The full covariance matrix $\mathrm{Var}(\mathbf{z})$ can be expressed readily using matrix notation:

$$\mathrm{Var}(\mathbf{z}) = E[(\mathbf{z} - E(\mathbf{z}))(\mathbf{z} - E(\mathbf{z}))'] \,.$$

3.3.1 Positive Definite Matrices

Definition 3.3.3: Positive definiteness

A real symmetric matrix $\mathbf{A}$ is

- Positive definiteness: $\mathbf{A} \succ 0 \iff \langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle > 0, \quad \forall \mathbf{x} \in \mathbb{R}^k \setminus \{0\}$.
- Positive semi–definiteness: $\mathbf{A} \succeq 0 \iff \langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^k$.

The scalar $\mathbf{x}'\mathbf{A}\mathbf{x}$ is the quadratic form associated with $\mathbf{A}$.

Example.

$\mathbf{I}_k$ is positive definite.

Theorem 3.3.4

If $\mathbf{B}$ (arbitrary matrix) is full-rank, then $\mathbf{B}'\mathbf{B}$ is positive definite.

Proof.

$$\mathbf{x}'\mathbf{B}'\mathbf{B}\mathbf{x} = (\mathbf{B}\mathbf{x})'(\mathbf{B}\mathbf{x}) = \|\mathbf{B}\mathbf{x}\|^2 \geq 0$$

□

Thus, the product $\mathbf{X}^\top\mathbf{X}$ is positive definite from the result $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ since we assume $\mathbf{X}$ is full-rank.

3.3.2 The OLS Covariance Matrix

We assume $\mathbf{X}$ is exogeneous $E(\mathbf{u} \mid \mathbf{X}) = 0$. Then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. Thus, we can rewrite

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = E\Big[(\hat{\boldsymbol{\beta}} - E(\hat{\boldsymbol{\beta}})(\hat{\boldsymbol{\beta}} - E(\hat{\boldsymbol{\beta}})'\Big] = E\Big[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\Big]$$

Assumption 3.3.1: Disturbance assumptions

Spherical disturbances on error covariance matrix:

$$\mathrm{Var}(\mathbf{u}) = E(\mathbf{u}\,\mathbf{u}') = \sigma^2\mathbf{I}_M = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma^2 \end{bmatrix}$$

This single statement implies

- $\mathrm{Var}(u_t) = \sigma^2 \quad \forall t \Rightarrow$ Homoskedasticity
- $\mathrm{Cov}(u_t, u_s) = 0 \quad \forall t \neq s \Rightarrow$ No autocorrelation

If there is Heteroskedasticity and/or Autocorrelation, OLS is still unbiased but no longer efficient; generalized least squares (GLS) or heteroskedasticity- and autocorrelation-consistent (HAC) methods are recommended.

Theorem 3.3.5

Under spherical disturbances and exogeneity, $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Recall $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}$.

Proof.

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = E\big[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\big] = E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\,\mathbf{u}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\big] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,E(\mathbf{u}\,\mathbf{u}')\,\mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}$$

where $E(\mathbf{u}\,\mathbf{u}') = E\big[(\mathbf{u} - E(\mathbf{u}))(\mathbf{u} - E(\mathbf{u}))'\big]$. Since $E(\mathbf{u}\,\mathbf{u}') = \mathrm{Var}(\mathbf{u}) = \sigma^2\mathbf{I}$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

□

Obviously, $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ is positive definite. It turns out we can do alternative proof of consistency using this theorem.

Corollary 3.3.6

Under spherical disturbances, $\left(\frac{\mathbf{X}'\mathbf{X}}{m}\right) \xrightarrow{p} \mathbf{S}$, and exogeneity, $\mathrm{Var}(\hat{\boldsymbol{\beta}}) \to 0$ which means $\hat{\boldsymbol{\beta}}$ is consistent.

Proof.

$$\mathrm{MSE}(\hat{\boldsymbol{\beta}}) = \underbrace{\mathrm{Bias}^2}_{0} + \mathrm{Var} = \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \frac{m}{m}\,\sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \frac{\sigma^2}{m}\left(\frac{\mathbf{X}'\mathbf{X}}{m}\right)^{-1}$$

Since $\left(\frac{\mathbf{X}'\mathbf{X}}{m}\right) \xrightarrow{p} \mathbf{S}$, then $\mathrm{Var}(\hat{\boldsymbol{\beta}}) \longrightarrow 0$. □

3.3.3 Multicollinearity

Consider $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} = \mathbf{x}_1\beta_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}$, where $\mathbf{x}_1$ is the column of interest and $\mathbf{X}_2$ contains the remaining $k-1$ columns. If $\mathbf{x}_1 \perp \mathbf{X}_2$, then $\mathbf{P}_2\mathbf{x}_1 = 0$, $\mathbf{M}_2\mathbf{x}_1 = \mathbf{x}_1$.
By FWL: $\mathbf{M}_2\mathbf{y} = \mathbf{M}_2\mathbf{x}_1\beta_1 + \mathbf{M_2}\mathbf{u}$, where $\mathbf{I} - \mathbf{M}_2 = \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2' = \mathbf{P}_2$. Then

$$\hat{\beta}_1 = (\mathbf{x}_1'\mathbf{M}_2\mathbf{x}_1)^{-1}\mathbf{x}_1'\mathbf{M}_2\mathbf{y} = \frac{\mathbf{x}_1'\mathbf{M}_2\mathbf{y}}{\mathbf{x}_1'\mathbf{M}_2\mathbf{x}_1} \overset{\mathbf{x}_1 \perp \mathbf{x}_2}{=} \frac{\mathbf{x}_1'\mathbf{y}}{\mathbf{x}_1'\mathbf{x}_1}$$

Under spherical errors,

$$\mathrm{Var}(\hat{\beta}_1) = \sigma^2(\mathbf{x}_1'\mathbf{M}_2\mathbf{x}_1)^{-1} = \frac{\sigma^2}{\mathbf{x}_1'\mathbf{M}_2\mathbf{x}_1} \overset{\mathbf{x}_1 \perp \mathbf{X}_2}{=} \frac{\sigma^2}{\mathbf{x}_1'\mathbf{x}_1}$$

Regress $\mathbf{x}_1$ on $\mathbf{X}_2$: $\mathbf{x}_1 = \mathbf{X}_2\boldsymbol{\lambda} + \mathbf{v}$, $\hat{\mathbf{v}} = \mathbf{M}_2\mathbf{x}_1$. Then

$$\|\hat{\mathbf{v}}\|^2 = \langle \hat{\mathbf{v}}, \hat{\mathbf{v}} \rangle = \hat{\mathbf{v}}'\hat{\mathbf{v}} = \mathbf{x}_1'\mathbf{M}_2'\mathbf{M}_2\mathbf{x}_1 = \mathbf{x}_1'\mathbf{M}_2\mathbf{x}_1 = SSR$$

Thus the variance of $\hat{\beta}_1$ is proportional to the inverse of the sum of squared residuals. When $\mathbf{x}_1$ is well explained by the other columns of $\mathbf{X}$, this SSR will be small, and the variance of $\hat{\beta}_1$ will consequently be large. When $\mathbf{x}_1$ is not well explained by the other columns of $\mathbf{X}$, this SSR will be large, and the variance of $\hat{\beta}_1$ will consequently be small.

Definition 3.3.7

The precision with which β_1 is estimated depends on $\mathbf{X}_2$ just as much as it depends on $\mathbf{x}_1$. If we then include some additional regressors, the estimate becomes much less precise. The reason for this is that the additional regressors do a much better job of explaining $\mathbf{x}_1$ than does a constant alone. As a consequence, the length of $\mathbf{M}_2\mathbf{x}_1$ is much less than the length of $\mathbf{M}_1\mathbf{x}_1$. This type of situation is sometimes referred to as collinearity, or **multicollinearity**, and the regressor $\mathbf{x}_1$ is said to be collinear with some of the other regressors.

Example.

Given the regressor matrix $\mathbf{X} = \left[\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\right]$, examine each column's dependence on the others by running k separate auxiliary regressions:

$$\begin{cases} \mathbf{X}_1 = \mathbf{X}_{(-1)}\boldsymbol{\lambda}_1 + \mathbf{v}_1, & R_1^2 = 0.05 \\ \mathbf{X}_2 = \mathbf{X}_{(-2)}\boldsymbol{\lambda}_2 + \mathbf{v}_2, & R_2^2 = 0.9 \\ \vdots & \\ \mathbf{X}_K = \mathbf{X}_{(-K)}\boldsymbol{\lambda}_K + \mathbf{v}_K & R_k^2 = 0.2, \end{cases}$$

3.3.4 Linear Functions of Parameter Estimates

Cobb–Douglas function: $y_t = A\,K_t^{\alpha}L_t^{\gamma}$ and $\alpha + \gamma = 1$ under CRS. Log-linear regression form:

$$\log y_t = \log A + \alpha \log K_t + \gamma \log L_t + u_t$$

where $\boldsymbol{\beta} = \begin{bmatrix} \log A \\ \alpha \\ \gamma \end{bmatrix}$. We know $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, but we want to know $\mathrm{Var}(\alpha + \gamma) = ?$. Set $\boldsymbol{\omega}' = [\,0\ 1\ 1\,]$. Then $\boldsymbol{\omega}'\boldsymbol{\beta} = \alpha + \gamma$. Then

$$E(\boldsymbol{\omega}'\hat{\boldsymbol{\beta}}) = \boldsymbol{\omega}'E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\omega}'\boldsymbol{\beta}$$

$$\mathrm{Var}(\boldsymbol{\omega}'\hat{\boldsymbol{\beta}}) = E\big[\boldsymbol{\omega}'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\boldsymbol{\omega}\big] = \boldsymbol{\omega}'E\big[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\big]\boldsymbol{\omega} = \sigma^2\,\boldsymbol{\omega}'(\mathbf{X}'\mathbf{X})^{-1}\,\boldsymbol{\omega}$$

The covariance matrix of $\hat{\beta}$ can be used to calculate the variance of any linear (strictly speaking, affine) function of $\hat{\beta}$. Suppose that we are interested in the variance of $\hat{\gamma}$, where $\gamma = \mathbf{w}^\top\beta$, $\hat{\gamma} = \mathbf{w}^\top\hat{\beta}$, and $\mathbf{w}$ is a k-vector of known coefficients. By choosing $\mathbf{w}$ appropriately, we can make γ equal to any one of the β_i, or to the sum of the β_i, or to any linear combination of the β_i in which we might be interested.

3.4 Efficiency of the OLS Estimator

Consider two **linear unbiased** estimators, $\hat{\boldsymbol{\beta}}$ (OLS) and some alternative $\tilde{\boldsymbol{\beta}}$.

Definition 3.4.1

$\hat{\boldsymbol{\beta}}$ is more efficient than $\tilde{\boldsymbol{\beta}}$ if $\Big[\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})\Big]$ is positive semi-definite.

Equivalently, for any vector $\boldsymbol{\omega} \in \mathbb{R}^d$:

$$\boldsymbol{\omega}'\Big[\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})\Big]\boldsymbol{\omega} \geq 0 \Leftrightarrow \boldsymbol{\omega}'\,\mathrm{Var}(\tilde{\boldsymbol{\beta}})\boldsymbol{\omega} - \boldsymbol{\omega}'\,\mathrm{Var}(\hat{\boldsymbol{\beta}})\boldsymbol{\omega} \geq 0 \Leftrightarrow \boldsymbol{\omega}'\,\mathrm{Var}(\tilde{\boldsymbol{\beta}})\boldsymbol{\omega} \;\geq\; \boldsymbol{\omega}'\,\mathrm{Var}(\hat{\boldsymbol{\beta}})\boldsymbol{\omega}$$

Recall $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. If $\tilde{\boldsymbol{\beta}}$ now denotes any linear estimator that is not the OLS estimator, we can always write

$$\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y} = ((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{C})\mathbf{y} = \hat{\boldsymbol{\beta}} + \mathbf{C}\mathbf{y}$$

Condition for unbiasedness

$$E(\tilde{\boldsymbol{\beta}}) = E(\hat{\boldsymbol{\beta}} + \mathbf{C}\mathbf{y}) = E(\hat{\boldsymbol{\beta}}) + E(\mathbf{C}\mathbf{y}) = \boldsymbol{\beta} + E(\mathbf{C}\mathbf{y})$$

Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$,

$$E(\mathbf{C}\mathbf{y}) = E\big(\mathbf{C}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})\big) = E[\mathbf{C}\mathbf{X}\boldsymbol{\beta}] + \underbrace{E(\mathbf{C}\mathbf{u})}_{=0}.$$

For $E(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, it must be that

$$\mathbf{C}\mathbf{X} = \mathbf{0}.$$

Theorem 3.4.2: Gauss–Markov theorem

If it is assumed that $\mathbf{X}$ is full rank, $\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$ and $\mathbb{E}(\mathbf{u}\mathbf{u}^\top) = \sigma^2\mathbf{I}$ in the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

then the OLS estimator $\hat{\boldsymbol{\beta}}$ is more efficient than any other linear unbiased estimator $\tilde{\boldsymbol{\beta}}$, namely Best Linear Unbiased Estimator (BLUE).

There are actually four assumptions!

Proof.

$$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) = \mathrm{Var}\big(\hat{\boldsymbol{\beta}} + (\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}})\big) = \mathrm{Var}(\hat{\boldsymbol{\beta}} + \mathbf{C}\mathbf{y}) = \mathrm{Var}(\hat{\boldsymbol{\beta}}) + \mathrm{Var}(\mathbf{C}\mathbf{y}) + \mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{C}\mathbf{y})$$

and

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{C}\mathbf{y}) = E\big[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\tilde{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}})'\big] = E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\,\mathbf{u}'\mathbf{C}'\big] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u}\mathbf{u}')\mathbf{C}' = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{C}' = \mathbf{0}$$

Then

$$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) = \mathrm{Var}(\hat{\boldsymbol{\beta}}) + \mathrm{Var}(\mathbf{C}\mathbf{y}) \Rightarrow \mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \mathrm{Var}(\mathbf{C}\mathbf{y})$$

We claim $\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite. Indeed, since $\mathbf{C}\mathbf{X} = \mathbf{0}$ and $\mathbf{C}\mathbf{C}' \geq 0$,

$$\mathrm{Var}(\mathbf{C}\mathbf{y}) = \mathrm{Var}\big(\mathbf{C}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})\big) = \mathrm{Var}(\mathbf{C}\mathbf{X}\boldsymbol{\beta} + \mathbf{C}\mathbf{u}) = \mathrm{Var}(\mathbf{C}\mathbf{u}) = E\big[\mathbf{C}\mathbf{u}\mathbf{u}'\mathbf{C}'\big] = \mathbf{C}\,E(\mathbf{u}\mathbf{u}')\,\mathbf{C}' = \sigma^2\mathbf{C}\mathbf{C}'$$

Thus, $\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})$ is positive semi-definite and then $\hat{\boldsymbol{\beta}}$ possesses the minimum variance. □

3.5 Residuals and Error Terms

Recall. Under $\left(\frac{\mathbf{X}'\mathbf{X}}{m}\right) \xrightarrow{p} \mathbf{S}$, spherical disturbances and exogeneity, we have $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.
In this section, we turn our attention to the statistical properties of $\hat{\mathbf{u}}$ as an estimator of $\mathbf{u}$. These properties are very important, because we will want to use $\hat{\mathbf{u}}$ for a number of purposes. In particular, we will want to use it to estimate σ^2, the variance of the error terms. We need an estimate of σ^2 if we

are to obtain an estimate of the covariance matrix of $\hat{\boldsymbol{\beta}}$.
From $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ and $E(\mathbf{u}\mathbf{u}') = \sigma^2\mathbf{I}$, we have

$$\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M_X}\,\mathbf{y} = \mathbf{M_X}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = \mathbf{M_X}\,\mathbf{u}$$

Theorem 3.5.1

$\hat{\mathbf{u}} \xrightarrow{P} \mathbf{u}$

Proof. Since $\hat{\boldsymbol{\beta}} \xrightarrow{P} \boldsymbol{\beta}$, then $\hat{\mathbf{u}} \xrightarrow{P} \mathbf{u}$ □

We will see that $\text{Var}(\hat{u}_t)$ is less than σ^2 for every observation. The easiest way to calculate the variance of $\hat{u}_t$ is to calculate the covariance matrix of the entire vector $\hat{\mathbf{u}}$:

$$\text{Var}(\hat{\mathbf{u}}) = \text{Var}(\mathbf{M_X}\mathbf{u}) = E\big[\mathbf{M_X}\mathbf{u}\,\mathbf{u}'\,\mathbf{M_X}\big] = \mathbf{M_X}E(\mathbf{u}\mathbf{u}')\mathbf{M_X} = \mathbf{M_X}\sigma^2\mathbf{I}\mathbf{M_X} = \sigma^2\mathbf{M_X}$$

Recall that H_t denotes the tth diagonal element of the projection matrix $\mathbf{P_X}$. Thus a typical diagonal element of $\mathbf{M_X}$ is $1 - H_t$. Therefore, it follows that

$$\text{Var}(\hat{u}_t) = \sigma^2(1 - H_t), \quad 0 \le H_t \le 1$$

Since $0 \le 1 - H_t < 1$,

$$\text{Var}(\hat{u}_t) = \sigma^2(1 - H_t) \le \sigma^2 = \text{Var}(u_t)$$

Just how much smaller will depend on H_t. It is clear that high-leverage observations, for which H_t is relatively large, will have residuals with smaller variance than low-leverage observations, for which H_t is relatively small.

3.5.1 Estimating the Variance of the Error Terms

Definition 3.5.2

Population moment condition $\text{Var}(u_t) = \sigma^2$.

Definition 3.5.3

Sample moment $\frac{1}{M}\sum_t u_t^2$.

The method of moments suggests that we can estimate σ^2 by using the corresponding sample moment. If we actually observed the u_t, this sample moment would be

$$\frac{1}{M}\sum_{t=1}^{M} u_t^2.$$

We do not observe the u_t, but we do observe the $\hat{u}_t$. Thus the simplest possible MM estimator is

$$\hat{\sigma}^2 \equiv \frac{1}{M}\sum_{t=1}^{M} \hat{u}_t^2.$$

This estimator is just the average of M squared residuals. It is consistent (Theorem 3.6.1). However, because each squared residual has expectation less than σ_0^2, , $\hat{\sigma}^2$ must be biased downward.

Theorem 3.5.4

Estimator $\hat{\sigma}^2 = \frac{1}{M}\sum_t \hat{u}_t^2$ is consistent yet biased downward.

Proof. Clearly from $\hat{u}_t \xrightarrow{p} u_t$. Since $\sum_t H_t = k$ and $\sum_t (1 - H_t) = M - k$, then

$$E(\hat{\sigma}^2) = \frac{1}{M}\sum_t E(\hat{u}_t^2) = \frac{1}{M}\sum_t \sigma^2(1 - H_t) = \frac{\sigma^2}{M}\sum_t (1 - H_t) = \sigma^2 \frac{M-k}{M} < \sigma^2.$$

□

The result suggests another MM estimator which will be unbiased:

$$s^2 = \frac{M}{M-k}\hat{\sigma}^2 = \frac{1}{M-k}\sum_t \hat{u}_t^2.$$

The only difference between $\hat{\sigma}^2$ and s^2 is that the former divides the SSR by M and the latter divides it by $M - k$. As a result, s^2 will be unbiased whenever $\hat{\boldsymbol{\beta}}$ is. Dividing by $M - k$ instead of by M eliminates this bias.
Virtually all OLS regression programs report s^2 as the estimated variance of the error terms. However, it is important to remember that, even though s^2 provides an unbiased estimate of σ^2, s itself does not provide an unbiased estimate of σ, because taking the square root of s^2 is a nonlinear operation.
If we replace σ^2 by s^2, we can obtain an unbiased estimate of $\mathrm{Var}(\hat{\boldsymbol{\beta}})$,

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = s^2(\mathbf{X}^\top \mathbf{X})^{-1}.$$

This is the standard estimate of the covariance matrix of the OLS parameter estimates under the assumption of IID errors.

3.6 Misspecification of Linear Regression Models

There are two types of misspecification:

- Over-specification (including irrelevant regressors)
  - Estimate: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{v}$
  - True: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$ with $\boldsymbol{\gamma} = \mathbf{0}$
  - $\hat{\boldsymbol{\beta}}$ stays unbiased and consistent, yet its variance is inflated, so efficiency falls.
- Under-specification (omitted variable bias)
  - Estimate: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$
  - True: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{v}$ with $\boldsymbol{\gamma} \neq \mathbf{0}$
  - If $\mathbf{Z}$ correlates with $\mathbf{X}$, then OLS coefficients on $\mathbf{X}$ become biased and inconsistent.

Note: for true models, all the assumptions we made in the previous sections hold.

3.6.1 Overspecification

- Estimate model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{v} \quad (\tilde{\boldsymbol{\beta}})$.
- True model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \mathbf{v} \quad (\boldsymbol{\gamma} = \mathbf{0})\ (\hat{\boldsymbol{\beta}})$.

Since $\mathbf{M}_z = \mathbf{I} - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$, by FWL,

$$\mathbf{M}_z\mathbf{y} = \mathbf{M}_z\mathbf{X}\boldsymbol{\beta} + \mathbf{M}_\mathbf{z}\mathbf{u}$$

We have

$$\tilde{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{M}_z\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_z\mathbf{y} \Rightarrow \mathrm{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{M}_z\mathbf{X})^{-1}$$

Since $\tilde{\boldsymbol{\beta}}$ is part of the OLS estimator of a correctly specified model, it should be unbiased. Indeed, if we replace $\mathbf{y}$ by $\mathbf{X}\boldsymbol{\beta}_0 + \mathbf{u}$, we find that

$$\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}_0 + (\mathbf{X}^\top\mathbf{M}_z\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{M}_z\mathbf{u}.$$

The conditional expectation of the second term on the right-hand side is 0, provided we take expectations conditional on $\mathbf{Z}$ as well as on $\mathbf{X}$.
For the true model,

$$\boldsymbol{\gamma} = \mathbf{0} \Rightarrow \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \Rightarrow \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

We claim $\hat{\boldsymbol{\beta}}$ is more efficient than $\tilde{\boldsymbol{\beta}}$. Now prove it in two cases:

- For $\beta_{1\times 1}$: because $\mathbf{M}_z$ is a contraction

  $$\mathbf{X}'\mathbf{M}_z\mathbf{X} = \|\mathbf{M}_z\mathbf{X}\|^2 \leq \|\mathbf{X}\|^2$$

  Then

  $$\mathrm{Var}(\tilde{\beta}) = \frac{\sigma^2}{\mathbf{X}'\mathbf{M}_z\mathbf{X}} = \sigma^2\big(\|\mathbf{M}_z\mathbf{X}\|^2\big)^{-1} \geq \sigma^2\big(\|\mathbf{X}\|^2\big)^{-1} = \mathrm{Var}(\hat{\beta}) \Rightarrow \mathrm{Var}(\tilde{\beta}) \geq \mathrm{Var}(\hat{\beta})$$

- In general case, for $\boldsymbol{\beta}_{k\times 1}$, we prove positive definiteness:

  $$\big[\mathrm{Var}(\hat{\boldsymbol{\beta}})\big]^{-1} - \big[\mathrm{Var}(\tilde{\boldsymbol{\beta}})\big]^{-1} = \big[\sigma^2(\mathbf{X}'\mathbf{X})^{-1}\big]^{-1} - \big[\sigma^2(\mathbf{X}'\mathbf{M}_z\mathbf{X})^{-1}\big]^{-1} = \frac{1}{\sigma^2}\mathbf{X}'\mathbf{X} - \frac{1}{\sigma^2}\big(\mathbf{X}'\mathbf{M}_z\mathbf{X}\big)$$

  $$= \frac{1}{\sigma^2}\mathbf{X}'(\mathbf{I} - \mathbf{M}_z)\mathbf{X} = \frac{1}{\sigma^2}\mathbf{X}'\mathbf{P}_z\mathbf{X} = \frac{1}{\sigma^2}\mathbf{X}'\mathbf{P}_z'\mathbf{P}_z\mathbf{X} = \frac{1}{\sigma^2}\big(\mathbf{X}'\mathbf{P}_z'\big)\big(\mathbf{P}_z\mathbf{X}\big) \geq 0$$

  Thus

  $$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}) \geq 0$$

  The last expression will be a zero matrix whenever $\mathbf{P}_z\mathbf{X} = \mathbf{O}$. This condition will hold when the two sets of regressors $\mathbf{X}$ and $\mathbf{Z}$ are mutually orthogonal, so that $\mathbf{Z}^\top\mathbf{X} = \mathbf{O}$. In this special case, $\tilde{\boldsymbol{\beta}}$ will be as efficient as $\hat{\boldsymbol{\beta}}$.

In general, however, including regressors that do not belong in a model will increase the variance of the estimates of the coefficients on the regressors that do belong, and the increase can be very great in many cases. The more other regressors there are, whether they truly belong in the model or not, the smaller will be this SSR, and, in consequence, the larger will be the variance of $\tilde{\boldsymbol{\beta}}$.

3.6.2 Under-specification

- Estimate model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$
- True model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_0 + \mathbf{Z}\boldsymbol{\gamma}_0 + \mathbf{v}$ with $\boldsymbol{\gamma}_0 \neq \mathbf{0}$

OLS breakdown

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta}_0 + \mathbf{Z}\boldsymbol{\gamma}_0 + \mathbf{v}) = \boldsymbol{\beta}_0 + \left(\tfrac{\mathbf{X}'\mathbf{X}}{M}\right)^{-1}\tfrac{\mathbf{X}'\mathbf{Z}}{M}\boldsymbol{\gamma}_0 + \left(\tfrac{\mathbf{X}'\mathbf{X}}{M}\right)^{-1}\tfrac{\mathbf{X}'\mathbf{v}}{M}$$

Theorem 3.6.1

$\hat{\boldsymbol{\beta}}$ is that it is now, in general, biased.

Proof. Since $E(\mathbf{v} \mid \mathbf{X}, \mathbf{Z}) = 0$, then

$$\mathrm{E}(\hat{\boldsymbol{\beta}}) = \mathrm{E}\big((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta}_0 + \mathbf{Z}\boldsymbol{\gamma}_0 + \mathbf{u})\big) = \boldsymbol{\beta}_0 + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0 + \mathrm{E}\big((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{u}\big)$$

$$= \boldsymbol{\beta}_0 + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0.$$

The second term in the last line will be equal to zero only when $\mathbf{X}^\top\mathbf{Z} = \mathbf{O}$ or $\boldsymbol{\gamma}_0 = \mathbf{0}$. Except in these very special cases, $\hat{\boldsymbol{\beta}}$ will generally be biased. □

Theorem 3.6.2

$\hat{\boldsymbol{\beta}}$ will generally be inconsistent.

Proof. Since $\frac{\mathbf{X}'\mathbf{X}}{M} \xrightarrow{p} \mathbf{S}_x$, $\frac{\mathbf{X}'\mathbf{Z}}{M} \xrightarrow{p} \mathbf{S}_{xz}$ and $\left(\frac{\mathbf{X}'\mathbf{X}}{M}\right)^{-1}\frac{\mathbf{X}'\mathbf{v}}{M} \xrightarrow{p} 0$,

$$\operatorname{plim}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \operatorname{plim}\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}\boldsymbol{\gamma}\big] = \operatorname{plim}\Big[\left(\frac{\mathbf{X}'\mathbf{X}}{M}\right)^{-1}\frac{\mathbf{X}'\mathbf{Z}}{M}\boldsymbol{\gamma}\Big] = \mathbf{S}_x^{-1}\mathbf{S}_{xz}\boldsymbol{\gamma} \neq \mathbf{0}$$

□

Since $\hat{\boldsymbol{\beta}}$ is biased, we cannot reasonably use its covariance matrix to evaluate its accuracy. Instead, we can use the mean squared error matrix, or MSE matrix, of $\hat{\boldsymbol{\beta}}$. This matrix is defined as

$$\operatorname{MSE}(\hat{\boldsymbol{\beta}}) \equiv \operatorname{E}\big((\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top\big).$$

The MSE matrix is equal to $\operatorname{Var}(\hat{\boldsymbol{\beta}})$ if $\hat{\boldsymbol{\beta}}$ is unbiased, but not otherwise. Recall

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0 = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0 + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{u}.$$

Therefore,

$$\begin{aligned}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0\boldsymbol{\gamma}_0^\top\mathbf{Z}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{u}\mathbf{u}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \\ &+ (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0\mathbf{u}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{u}\boldsymbol{\gamma}_0^\top\mathbf{Z}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}.\end{aligned}$$

The second term here has expectation $\sigma_0^2(\mathbf{X}^\top\mathbf{X})^{-1}$, and the third and fourth terms, one of which is the transpose of the other, have expectation zero. Thus we conclude that

$$\operatorname{MSE}(\hat{\boldsymbol{\beta}}) = \sigma_0^2(\mathbf{X}^\top\mathbf{X})^{-1} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Z}\boldsymbol{\gamma}_0\boldsymbol{\gamma}_0^\top\mathbf{Z}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \operatorname{Var} + \operatorname{Bias}^2 \tag{3.4}$$

The first term is what the covariance matrix would be if we were estimating a correctly specified model, and the second term arises from the bias of $\hat{\boldsymbol{\beta}}$.
By FWL: $\mathbf{M}_z\mathbf{y} = \mathbf{M}_z\mathbf{X}\boldsymbol{\beta}_0 + \mathbf{M}_\mathbf{z}\mathbf{u}$, then

$$\operatorname{Var}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{M}_z\mathbf{X})^{-1} \tag{3.5}$$

We would like to compare $\operatorname{MSE}(\hat{\boldsymbol{\beta}})$, expression (3.4), with $\operatorname{MSE}(\tilde{\boldsymbol{\beta}}) = \operatorname{Var}(\tilde{\boldsymbol{\beta}})$, which is given by expression (3.5). However, no unambiguous comparison is possible. The first term in (3.4) cannot be larger, in the matrix sense, than (3.5). Thus, if the bias is small, the second term will be small, and it may well be that $\hat{\boldsymbol{\beta}}$ is more efficient than $\tilde{\boldsymbol{\beta}}$. However, if the bias is large, the second term will necessarily be large, and $\hat{\boldsymbol{\beta}}$ will be less efficient than $\tilde{\boldsymbol{\beta}}$. Of course, it is quite possible that some parameters may be estimated more efficiently by $\hat{\boldsymbol{\beta}}$ and others more efficiently by $\tilde{\boldsymbol{\beta}}$.
It is tempting to conclude from this discussion that underspecification is a much more severe problem than overspecification. After all, the former constitutes misspecification, but the latter does not. In consequence, as we have seen, underspecification leads to biased estimates and an estimated covariance matrix that may be severely misleading, while overspecification merely leads to inefficiency.

Chapter 4

Hypothesis Testing and Confidence Intervals

4.1 Review of Distributions

Theorem 4.1.1

If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$, where $\mathbf{z}$ is an $m \times 1$ vector, and $\mathbf{x} = \boldsymbol{\mu} + \mathbf{A}\mathbf{z}$, then $\mathbf{x} \sim N(\boldsymbol{\mu}, \mathbf{A}\mathbf{A}')$.

Definition 4.1.2: Chi-square

If $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$, then $y = \|\mathbf{z}\|^2 = \mathbf{z}'\mathbf{z} = \sum_{t=1}^{m} z_t^2 \sim \chi^2(m)$

Theorem 4.1.3

If $y \sim \chi^2(m)$, then $E(y) = m$ and $\operatorname{Var}(y) = 2m$.

Theorem 4.1.4

If $y_1 \sim \chi^2(m_1)$, $y_2 \sim \chi^2(m_2)$, and y_1, y_2 are independent, then $y = y_1 + y_2 \sim \chi^2(m_1 + m_2)$

Theorem 4.1.5

If $\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Omega})$, where $\mathbf{X}$ is an $m \times 1$ vector, then $\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X} \sim \chi^2(m)$.

Theorem 4.1.6

If $\operatorname{rank}(\mathbf{P}) = r$ and $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$, then $\mathbf{z}'\mathbf{P}\mathbf{z} \sim \chi^2(r)$.

Definition 4.1.7: t-distribution

Let $z \sim N(0, 1)$ and $y_1 \sim \chi^2(m)$, which are both scalar. Then

$$t = \frac{z}{\sqrt{y_1/m}} \sim t(m) \xrightarrow[m \to \infty]{} N(0, 1)$$

$t(1)$: cauchy distribution. It has no moments at all.

Definition 4.1.8: Fisher distribution

Let $y_1 \sim \chi^2(m_1)$ and $y_2 \sim \chi^2(m_2)$. Then $F = \frac{y_1/m_1}{y_2/m_2} \sim F(m_1, m_2)$.

Theorem 4.1.9

- $F(1, m_2) = \big[t(m_2)\big]^2$
- $F(m_1, m_2) : \; m_1 F(m_1, m_2) \xrightarrow[m_2 \to \infty]{} \chi^2(m_1)$

4.2 Tests of a Single Restriction

Consider the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad \mathbf{u} \sim \mathcal{N}\big(\mathbf{0},\, \sigma^2\mathbf{I}\big)\,,$$

where $\mathbf{X}$ is an $m \times k$ matrix of regressors, so that there are m observations and k regressors, and it is assumed that the error vector $\mathbf{u}$ is statistically independent of the matrix $\mathbf{X}$. Note $\mathbf{u} \sim \mathcal{N}\big(\mathbf{0},\, \sigma^2\mathbf{I}\big)$ is our new assumption!
Let us partition $\boldsymbol{\beta}$ as $[\boldsymbol{\beta}_1 : \; \beta_2]$, where $\boldsymbol{\beta}_1$ is a $(k-1)$-vector and β_2 is a scalar, and consider

$$\begin{cases} H_0 : \beta_2 = 0 \\ H_1 : \beta_2 \neq 0 \end{cases}$$

When $\mathbf{X}$ is partitioned conformably with $\boldsymbol{\beta}$, the model above can be rewritten as

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \beta_2\mathbf{x}_2 + \mathbf{u}, \qquad \mathbf{u} \sim \mathcal{N}\big(\mathbf{0},\, \sigma^2\mathbf{I}\big)\,,$$

where $\mathbf{X}_1$ denotes an $m \times (k-1)$ matrix and $\mathbf{x}_2$ denotes an n-vector, with $\mathbf{X} = [\,\mathbf{X}_1 \;\; \mathbf{x}_2\,]$.
By the FWL Theorem, $\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\beta_2 + \mathbf{M_1}\mathbf{u}$, we have

$$\hat{\beta}_2 = \big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y} = \frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{y}}{\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2} \quad \text{and} \quad \mathrm{Var}(\hat{\beta}_2) = \sigma^2\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{-1} = \frac{\sigma^2}{\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2}.$$

Under the unrealistic assumption that σ is known, the test statistic is

$$Z = \frac{\hat{\beta}_2 - \beta_2^0}{\sqrt{\mathrm{Var}(\hat{\beta}_2)}} = \frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{y}}{\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2}\left(\frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2}{\sigma^2}\right)^{1/2} = \frac{1}{\sigma}\,\frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{y}}{\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{1/2}}$$

If the data are actually generated by the model with $\beta_2 = 0$, then

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}) = \mathbf{M}_1\mathbf{u}$$

Therefore,

$$z = \frac{1}{\sigma}\,\frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{y}}{\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{1/2}} = \frac{1}{\sigma}\,\frac{\mathbf{X}_2'\mathbf{M}_1\mathbf{u}}{\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{1/2}}$$

It is now easy to see that $z \sim N(0,1)$. Since $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, then

$$\mathbf{X}_2'\mathbf{M}_1\mathbf{u} \sim N\big(0, \sigma^2\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big) \quad \Rightarrow \quad z \sim N(0,1).$$

In order to handle the more realistic case in which we do not know the variance of the error terms, we need to replace σ by s.

$$s^2 = \frac{1}{M-k}\sum_t \hat{u}_t^2 = \frac{1}{M-k}\,\hat{\mathbf{u}}^\top\hat{\mathbf{u}} = \frac{1}{M-k}\,\mathbf{y}'\mathbf{M}_X\mathbf{y}.$$

So we obtain the test statistic

$$t \equiv \frac{\mathbf{x}_2^\top \mathbf{M}_1 \mathbf{y}}{s\,(\mathbf{x}_2^\top \mathbf{M}_1 \mathbf{x}_2)^{1/2}} = \left(\frac{\mathbf{y}^\top \mathbf{M}_X \mathbf{y}}{m-k}\right)^{-1/2} \frac{\mathbf{x}_2^\top \mathbf{M}_1 \mathbf{y}}{(\mathbf{x}_2^\top \mathbf{M}_1 \mathbf{x}_2)^{1/2}}.$$

As we will now demonstrate, this test statistic is distributed as $t(m-k)$ under the null hypothesis. Not surprisingly, it is called a t statistic. The t statistic defined above can be rewritten as

$$t = \left(\frac{\mathbf{y}'\mathbf{M}_X \mathbf{y}}{(m-k)\sigma^2}\right)^{-1/2} \frac{\mathbf{X}_2'\mathbf{M}_1 \mathbf{y}}{\sigma\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{1/2}} = \frac{z}{\left(\mathbf{y}^\top \mathbf{M}_X \mathbf{y}/((m-k)\sigma^2)\right)^{1/2}},$$

which has the form of such a ratio. We have already shown that $z \sim N(0,1)$. Thus it only remains to show that $\mathbf{y}^\top \mathbf{M}_X \mathbf{y}/\sigma^2 \sim \chi^2(m-k)$ and that the random variables in the numerator and denominator are independent.
Because $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$ then $\boldsymbol{\epsilon} = \frac{\mathbf{u}}{\sigma} \sim N(\mathbf{0}, \mathbf{I})$ and

$$\frac{\mathbf{y}'\mathbf{M}_X \mathbf{y}}{\sigma^2} = \frac{\mathbf{u}'\mathbf{M}_X \mathbf{u}}{\sigma^2} = \boldsymbol{\epsilon}'\mathbf{M}_X \boldsymbol{\epsilon} \sim \chi^2(m-k), \text{ with } \operatorname{rank}(\mathbf{M}_X) = m-k$$

We claim z and $\boldsymbol{\epsilon}'\mathbf{M}_X\boldsymbol{\epsilon}$ are independent (we omit the proof here). Thus,

$$t = \frac{z}{\sqrt{\dfrac{\mathbf{y}'\mathbf{M}_X \mathbf{y}}{(m-k)\sigma^2}}} \sim t(m-k) \xrightarrow[m\to\infty]{} N(0,1)$$

The above results hold under $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$. However, with $\mathbf{u} \sim \text{i.i.d}(\mathbf{0}, \sigma^2\mathbf{I})$, the same conclusions hold asymptotically by the Central Limit Theorem.

Theorem 4.2.1: Central Limit Theorem

For i.i.d $x_t \sim (\mu, \sigma^2)$, $\bar{Z} = \frac{1}{\sqrt{m}} \sum_t \frac{x_t - \mu}{\sigma} \xrightarrow{d} N(0,1)$.

4.3 Tests of Several Restrictions

Let us suppose that there are r restrictions, with $r \leq k$. As before, we consider

$$\begin{cases} H_0 : \beta_2 = 0 \\ H_1 : \beta_2 \neq 0 \end{cases}$$

The alternative hypothesis is

$$H_1 : \quad \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}, \qquad \mathbf{u} \sim N(0, \sigma^2\mathbf{I}).$$

Here $\mathbf{X}_1$ is an $m \times k_1$ matrix, $\mathbf{X}_2$ is an $m \times k_2$ matrix, $\boldsymbol{\beta}_1$ is a k_1-vector, $\boldsymbol{\beta}_2$ is a k_2-vector, $k = k_1 + k_2$, and the number of restrictions $r = k_2$. The null hypothesis is the regression model

$$H_0 : \quad \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{v}, \qquad \mathbf{v} \sim N(0, \sigma^2\mathbf{I}),$$

in which we impose the restriction that $\boldsymbol{\beta}_2 = \mathbf{0}$. Put together,

$$\begin{cases} H_0 : \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{v} & \text{Restricted model} \\ H_1 : \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u} & \text{Unrestricted model} \end{cases}$$

As we saw before, the restricted model must always fit worse than the unrestricted model, in the sense that the SSR from the restricted model cannot be smaller, and will almost always be larger, than the SSR from the unrestricted model. Therefore, it seems natural to base a test statistic on the difference between these two SSRs. If USSR denotes the unrestricted sum of squared residuals, and RSSR denotes

the restricted sum of squared residuals, the appropriate test statistic is

$$F \equiv \frac{(\text{RSSR} - \text{USSR})/r}{\text{USSR}/(m-k)}.$$

Under the null hypothesis, this test statistic follows the F distribution with r and $m-k$ degrees of freedom. Not surprisingly, it is called an F statistic.
It is easy to see

$$\text{RSSR} = \hat{\mathbf{v}}'\hat{\mathbf{v}} = \mathbf{y}'\mathbf{M}_1\mathbf{y}, \qquad \text{USSR} = \hat{\mathbf{u}}'\hat{\mathbf{u}} = \mathbf{y}'\mathbf{M}_X\mathbf{y}$$

By FWL theorem,

$$\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}.$$

So $\text{TSS} = \mathbf{y}'\mathbf{M}_1'\mathbf{M}_1\mathbf{y} = \mathbf{y}'\mathbf{M}_1\mathbf{y} = \text{RSSR}$ and $\text{SSR} = \hat{\mathbf{u}}'\hat{\mathbf{u}} = \text{USSR}$. Then

$$\text{TSS} = \text{ESS} + \text{SSR} \Rightarrow \text{RSSR} = \text{ESS} + \text{USSR} \Rightarrow \text{RSSR} - \text{USSR} = \text{ESS}$$

The fitted values are

$$\hat{\beta}_2 = (\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y}$$

Then

$$\mathbf{M}_1\mathbf{X}_2\hat{\beta}_2 = \mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y}$$

Thus,

$$\begin{aligned}\text{RSSR} - \text{USSR} = \text{ESS} &= \mathbf{y}'\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y} \\ &= (\mathbf{M}_1\mathbf{X}_2\hat{\beta}_2)'(\mathbf{M}_1\mathbf{X}_2\hat{\beta}_2) = \mathbf{y}'\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y}.\end{aligned}$$

The F statistic can be written as

$$F = \frac{\mathbf{y}^\top\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{M}_1\mathbf{y}/r}{\mathbf{y}^\top\mathbf{M}_X\mathbf{y}/(m-k)}.$$

Under $H_0 : \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{v}$ gives $\mathbf{M}_1\mathbf{y} = \mathbf{M}_1\mathbf{v}$. And under $H_1 : \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}$ gives $\mathbf{M}_X\mathbf{y} = \mathbf{M}_X\mathbf{u}$. Thus, the F statistic reduces to

$$F = \frac{\mathbf{u}'\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{u}\ /r}{\mathbf{u}'\mathbf{M}_X\mathbf{u}/(m-k)}\frac{\sigma^2}{\sigma^2} = \frac{\boldsymbol{\varepsilon}'\mathbf{M}_1\mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{M}_1\boldsymbol{\varepsilon}\ /r}{\boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon}/(m-k)}$$

where $\boldsymbol{\varepsilon} = \frac{\mathbf{u}}{\sigma} \sim N(\mathbf{0}, \mathbf{I})$. Since $\text{rank}(\mathbf{M}_X) = m-k$, we have

$$\boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon} \sim \chi^2(m-k),.$$

Set $\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2} = \mathbf{M}_1\mathbf{X}_2\big(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\big)^{-1}\mathbf{X}_2'\mathbf{M}_1$ with $\text{Rank}(\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}) = r$. Then

$$\boldsymbol{\varepsilon}'\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}\boldsymbol{\varepsilon} \sim \chi^2(r)$$

Moreover, the random variables in the numerator and denominator are independent, because $\mathbf{M}_X$ and $\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}$ project onto mutually orthogonal subspaces:

$$\mathbf{M}_\mathbf{X}\mathbf{M}_1\mathbf{X}_2 = (\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{M}_1\mathbf{X}_2 = \mathbf{M}_1\mathbf{X}_2 - \mathbf{P}_\mathbf{X}\mathbf{M}_1\mathbf{X}_2 = \mathbf{M}_1\mathbf{X}_2 - \mathbf{M}_1\mathbf{X}_2 = 0.$$

Thus,

$$F = \frac{\boldsymbol{\varepsilon}'\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}\boldsymbol{\varepsilon}/r}{\boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon}/(m-k)} = \frac{\chi_r^2/r}{\chi_{m-k}^2/(m-k)} \sim F(r,\, m-k)$$

Note that

$$rF(r,\, m-k) \xrightarrow[m\to\infty]{} \chi^2(r)$$

Chapter 5

Generalized Least Squares

5.1 Heteroskedasticity and Autocorrelation

In the previous chapter, we have

$$\mathbb{E}(\mathbf{u}\mathbf{u}') = \sigma^2\mathbf{I} \;\to\; \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \;\to\; s^2(\mathbf{X}'\mathbf{X})^{-1},$$

that is we assume spherical disturbances and use standard error to infer covariance. However, spherical disturbances is not always correct and so the estimated covariance is wrong. In this chapter, we relax the spherical-errors assumption $E(\mathbf{u}\mathbf{u}') = \sigma^2\mathbf{I}$ to $E(\mathbf{u}\mathbf{u}') = \mathbf{\Omega}$, where $\mathbf{\Omega}$,

$$\mathbf{\Omega} = \begin{bmatrix} \omega_{11} & \omega_{12} & \cdots & \omega_{1M} \\ \omega_{21} & \omega_{22} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{M1} & \cdots & \cdots & \omega_{MM} \end{bmatrix}$$

the covariance matrix of the error terms, is a positive definite $n \times n$ matrix. We will be concerned with the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbb{E}(\mathbf{u}\mathbf{u}^\top) = \mathbf{\Omega}. \tag{5.1}$$

Then the OLS estimator is

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

It is unbiased:

$$E(\hat{\boldsymbol{\beta}}_{\text{OLS}}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u} \mid \mathbf{X}) = \boldsymbol{\beta}.$$

It is also consistent:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} = \left(\frac{\mathbf{X}'\mathbf{X}}{M}\right)^{-1}\frac{\mathbf{X}'\mathbf{u}}{M} \xrightarrow{p} 0.$$

Whatever the form of the error covariance matrix $\mathbf{\Omega}$, the covariance matrix of the OLS estimator is

$$\text{Var}(\hat{\boldsymbol{\beta}}_{\text{OLS}}) = E\big[(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})'\big]$$

$$= E\big[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\,\mathbf{u}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\big] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u}\mathbf{u}')\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbf{\Omega}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}.$$

This form of covariance matrix is often called a sandwich covariance matrix. Two special cases:

- When $\mathbf{\Omega}$ is

$$\mathbf{\Omega} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_m^2 \end{bmatrix}$$

then the error terms are still uncorrelated, but they are heteroskedastic. In this case,

$$\mathrm{Var}(u_t) = \sigma_t^2, \qquad E(u_t u_{t-s}) = 0$$

we need robust to conduct valid inference.

- When $\mathbf{\Omega}$ is

$$\mathbf{\Omega} = \begin{bmatrix} \sigma^2 & \rho & \rho^2 & \cdots & \rho^{m-1} \\ \rho & \sigma^2 & \rho & \cdots & \vdots \\ \rho^2 & \rho & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \sigma^2 & \rho \\ \rho^{m-1} & \cdots & \cdots & \rho & \sigma^2 \end{bmatrix}$$

It is called Autocorrelation. In this case,

$$\mathrm{Var}(u_t) = \sigma^2, \qquad E(u_t u_{t-s}) = \rho^s$$

For general case

$$\mathbf{\Omega} = \begin{bmatrix} \omega_{11} & \omega_{12} & \cdots & \omega_{1M} \\ \omega_{21} & \omega_{22} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{M1} & \cdots & \cdots & \omega_{MM} \end{bmatrix},$$

if we knew the w_{ij}, we could easily evaluate the sandwich covariance matrix. But it is assumed that we do not know the w_{ij}. Full estimation of all $\sum_{t=1}^{M} t = \frac{M(M+1)}{2}$ elements is infeasible.
For Heteroskedastic $\mathbf{\Omega}$, White provide feasible, consistent variance estimates without specifying the full error structure.

Theorem 5.1.1: White heteroskedasticity-robust estimator

For $\mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbf{\Omega}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, where $\mathbf{\Omega} = \mathrm{diag}(\omega_1^2, \ldots, \omega_n^2)$

$$(\mathbf{X}'\mathbf{\Omega}\,\mathbf{X})_{ij} \;=\; \Big[\sum_t \omega_t^2\, x_{ti}\, x_{tj}\Big] \;=\; [\,a_{ij}\,] \simeq \Big[\sum_t \hat{u}_t^2\, x_{ti}\, x_{tj}\Big]$$

where $\omega_t^2 \;\leftrightarrow\; \hat{u}_t^2$

5.2 The GLS Estimator

Suppose $\mathbf{\Omega}$ is known. Since $\mathbf{\Omega}$ is postive definite, we can always find $\mathbf{\Psi}$ such that $\mathbf{\Omega}^{-1} = \mathbf{\Psi}\,\mathbf{\Psi}'$ (Cholesky Factorization). Note that the matrix $\boldsymbol{\psi}$ must be invertible ($\boldsymbol{\psi}' = \boldsymbol{\psi}^{-1}$). Then we can transform the model from (5.1) to

$$\mathbf{\Psi}'\mathbf{y} = \mathbf{\Psi}'\mathbf{X}\boldsymbol{\beta} + \mathbf{\Psi}'\mathbf{u}$$

and

$$E\big[\mathbf{\Psi}'\mathbf{u}\,\mathbf{u}'\mathbf{\Psi}\big] = \mathbf{\Psi}'E(\mathbf{u}\mathbf{u}')\mathbf{\Psi} = \mathbf{\Psi}'\mathbf{\Omega}\mathbf{\Psi} = \mathbf{\Psi}'\big(\mathbf{\Psi}'^{-1}\mathbf{\Psi}^{-1}\big)\mathbf{\Psi} = \mathbf{I}$$

which means the transformed errors are spherical. The OLS estimator of $\boldsymbol{\beta}$ from this regression is

$$\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = \big(\mathbf{X}'\mathbf{\Psi}\mathbf{\Psi}'\mathbf{X}\big)^{-1}\mathbf{X}'\mathbf{\Psi}\mathbf{\Psi}'\mathbf{y} = \big(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X}\big)^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}.$$

This estimator is called the generalized least squares, or GLS, estimator of $\boldsymbol{\beta}$. The variance of GLS estimator is

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}) = \big(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X}\big)^{-1}$$

Compare with $\mathrm{Var}(\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, the GLS estimator $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}}$ will in most cases be more efficient, and will never be less efficient, than the OLS estimator $\hat{\boldsymbol{\beta}}$. Thus, GLS achieves the minimum variance among all linear unbiased estimators.

5.3 Weighted Least Squares

It is particularly easy to obtain GLS estimates when the error terms are heteroskedastic but uncorrelated. This implies that the matrix $\mathbf{\Omega}$ is diagonal.

$$\mathbf{\Omega} = \begin{bmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \omega_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_m^2 \end{bmatrix}$$

Since $\mathbf{\Omega}^{-1} = \mathbf{\Psi}\mathbf{\Psi}'$, we can find its inverse

$$\mathbf{\Omega}^{-1} = \begin{bmatrix} \omega_1^{-2} & 0 & \cdots & 0 \\ 0 & \omega_2^{-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_m^{-2} \end{bmatrix}, \text{ where } \mathbf{\Psi}' = \begin{bmatrix} \omega_1^{-1} & 0 & \cdots & 0 \\ 0 & \omega_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_m^{-1} \end{bmatrix}.$$

Premultiply the model:

$$\mathbf{\Psi}'\mathbf{y} = \mathbf{\Psi}'\mathbf{X}\boldsymbol{\beta} + \mathbf{\Psi}'\mathbf{u}.$$

Thus we see that, for a typical observation, regression can be written as

$$\omega_t^{-1} y_t = \omega_t^{-1}\mathbf{X}_t\boldsymbol{\beta} + \omega_t^{-1} u_t$$

This regression is to be estimated by OLS. The regressand and regressors are simply the dependent and independent variables multiplied by ω_t^{-1}, and the variance of the error term is clearly 1:

$$\mathrm{Var}(u_t^*) = \mathrm{Var}\left(\frac{1}{\omega_t} u_t\right) = \frac{1}{\omega_t^2}\,\mathrm{Var}(u_t) = \frac{1}{\omega_t^2} \cdot \omega_t^2 = 1.$$

For obvious reasons, this special case of GLS estimation is often called weighted least squares, or WLS. The weight given to each observation when we run regression is ω_t^{-1}. Observations for which the variance of the error term is large are given low weights, and observations for which it is small are given high weights.

5.4 Feasible Generalized Least Squares

In practice, the covariance matrix $\mathbf{\Omega}$ is often not known even up to a scalar factor. However, in many cases it is reasonable to suppose that $\mathbf{\Omega}$ depends in a known way on a vector of unknown parameters $\boldsymbol{\gamma}$, denoted by $\mathbf{\Omega}(\boldsymbol{\gamma}) = \mathbf{\Omega}$. If so, it may be possible to estimate $\boldsymbol{\gamma}$ consistently, so as to obtain $\mathbf{\Omega}(\hat{\boldsymbol{\gamma}})$, say. Then $\mathbf{\Psi}(\hat{\boldsymbol{\gamma}})$ can be defined as $\mathbf{\Omega}^{-1} = \mathbf{\Psi}\,\mathbf{\Psi}'$, and GLS estimates computed conditional on $\mathbf{\Psi}(\hat{\boldsymbol{\gamma}})$. This type of procedure is called feasible generalized least squares, or feasible GLS, because it is feasible in many cases when ordinary GLS is not.
As a simple example, suppose we want to obtain feasible GLS estimates of the linear regression model:

$$y_t = \mathbf{X}_t\boldsymbol{\beta} + u_t, \quad \mathrm{Var}(u_t) = \sigma_t^2 = \exp(\mathbf{z}_t\gamma),$$

1. Start by computing OLS estimates $\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} \xrightarrow{p} \boldsymbol{\beta}$. This allows us to calculate a vector of OLS residuals with typical element $\hat{u}_t \xrightarrow{p} u_t$.

2. Then run the auxiliary linear regression $\log \hat{u}_t^2 = \mathbf{z}_t\boldsymbol{\gamma} + v_t$ to find the OLS estimates $\hat{\boldsymbol{\gamma}} \xrightarrow{p} \boldsymbol{\gamma}$.

3. These estimates are then used to compute $\hat{\omega}_t = \sqrt{\exp(\mathbf{z}_t\hat{\boldsymbol{\gamma}})} \xrightarrow{p} \omega_t$.

4. Finally, feasible GLS estimates of $\boldsymbol{\beta}$ are obtained by using ordinary least squares to estimate regression, with the estimates $\hat{\omega}_t$ replacing the unknown ω_t.

5.5 Panel Data

Many data sets are measured across two dimensions. One dimension is time, and the other is usually called the cross-section dimension. Data of this type are often referred to as panel data.
For simplicity, we restrict our attention to the linear regression model

$$y_{it} = \mathbf{X}_{it}\boldsymbol{\beta} + u_{it}, \quad i = 1, \ldots, M,\ t = 1, \ldots, T$$

where $\mathbf{X}_{it}$ is a $1 \times k$ vector of observations on explanatory variables. There are assumed to be M cross-sectional units and T time periods, for a total of $n = MT$ observations. Visually,

$$\mathbf{y}_{(MT\times 1)} = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1T} \\ y_{21} \\ \vdots \\ y_{2T} \\ \vdots \\ y_{M1} \\ \vdots \\ y_{MT} \end{bmatrix}, \quad \mathbf{X}_{(MT\times k)} = \begin{bmatrix} x'_{11} \\ \vdots \\ x'_{1T} \\ x'_{21} \\ \vdots \\ x'_{2T} \\ \vdots \\ x'_{M1} \\ \vdots \\ x'_{MT} \end{bmatrix}, \quad \mathbf{U}_{(MT\times 1)} = \begin{bmatrix} u_{11} \\ \vdots \\ u_{1T} \\ u_{21} \\ \vdots \\ u_{2T} \\ \vdots \\ u_{M1} \\ \vdots \\ u_{MT} \end{bmatrix}$$

Under conditions: $\mathrm{E}\big(v_{it} \,\big|\, \mathbf{X}_{it}\big) = 0$, $\mathrm{E}\big(v_{it}^2\big) = \sigma_u^2$, and $\boldsymbol{\Omega} = \sigma^2\mathbf{I}$, OLS is BLUE.

5.5.1 Error-Components Models

The two most popular approaches for dealing with panel data are both based on what are called error-components models. The idea is to specify the error term u_{it} in as

$$u_{it} = v_i + \varepsilon_{it} + (\xi_t)$$

Here v_i affects all observations for cross-sectional unit i, ε_{it} affects only observation it, and ξ_t affects all observations for time period t (we omit it here). We assume:

$$\mathrm{E}\big(\varepsilon_{it} \,\big|\, \mathbf{X}_{it}\big) = 0, \quad \mathrm{E}(\varepsilon_{it}\varepsilon_{is}) = 0, \quad \mathrm{E}(v_i\varepsilon_{it}) = 0.$$

We also know $\mathrm{Var}(u_{it}) = \sigma_v^2 + \sigma_\varepsilon^2 = \sigma_u^2$. Then

$$\mathrm{E}(u_{it}u_{is}) = \mathrm{E}\big[(v_i + \varepsilon_{it})(v_i + \varepsilon_{is})\big] = \mathrm{E}\big[v_i^2 + \varepsilon_{it}v_i + v_i\varepsilon_{is} + \varepsilon_{it}\varepsilon_{is}\big] = \mathrm{E}\big[v_i^2\big] = \sigma_v^2$$

If the data are ordered by the cross-sectional units in M blocks of T observations each, this matrix has the form

$$\boldsymbol{\Omega} = \begin{bmatrix} \boldsymbol{\Sigma} & 0 & \cdots & 0 \\ 0 & \boldsymbol{\Sigma} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{\Sigma} \end{bmatrix}, \quad \text{and } \boldsymbol{\Sigma}_{T\times T} = \begin{bmatrix} \sigma_v^2 + \sigma_\varepsilon^2 & \sigma_v^2 & \cdots & \sigma_v^2 \\ \sigma_v^2 & \sigma_v^2 + \sigma_\varepsilon^2 & \cdots & \sigma_v^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_v^2 & \sigma_v^2 & \cdots & \sigma_v^2 + \sigma_\varepsilon^2 \end{bmatrix}$$

Then the robust OLS variance is

$$\mathrm{Var}\big(\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}\big) = \big(\mathbf{X}'\mathbf{X}\big)^{-1}\mathbf{X}'\,\hat{\boldsymbol{\Omega}}\,\mathbf{X}\big(\mathbf{X}'\mathbf{X}\big)^{-1}$$

where $\hat{\boldsymbol{\Omega}}$ is built from the estimates $\big(\hat{\sigma}_v^2,\, \hat{\sigma}_\varepsilon^2\big)$. There are two scenarios:

- If the individual effect is exogenous, $\mathrm{E}(v_i \mid \mathbf{X}_{it}) = 0$, GLS equals the random-effects estimator and is efficient $\hat{\boldsymbol{\beta}}_{\mathrm{GLS}} = \hat{\boldsymbol{\beta}}_{\mathrm{RE}}$.
- If that exogeneity fails $\mathrm{E}(v_i \mid \mathbf{X}_{it}) \neq 0$, RE/GLS is biased; switch to fixed-effects or other consistent methods.

5.5.2 Fixed-Effects Estimation

If $\mathrm{E}\big(v_i \,\big|\, \mathbf{X}_{it}\big) \neq 0$, then

$$\mathrm{E}\big(u_{it} \mid \mathbf{X}_{it}\big) = \mathrm{E}\Big[\, v_i + \varepsilon_{it} \,\Big|\, \mathbf{X}_{it}\Big] = \mathrm{E}\big(v_i \mid \mathbf{X}_{it}\big) + \mathrm{E}\big(\varepsilon_{it} \mid \mathbf{X}_{it}\big) = \mathrm{E}\big(v_i \mid \mathbf{X}_{it}\big) \neq 0$$

Thus, OLS is not consistent. We can solve this using fixed- effects estimation. The model that underlies fixed-effects estimation can be written as follows:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{V} + \boldsymbol{\varepsilon} \;=\; \mathbf{X}\boldsymbol{\beta} + \mathbf{D}\boldsymbol{\eta} + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ are n-vectors with typical elements y_{it} and ε_{it}, respectively, and $\mathbf{D}$ is an $n \times M$ matrix of dummy variables, constructed in such a way that the element in the row corresponding to observation it, for $i = 1, \ldots, M$ and $t = 1, \ldots, T$, and column j, for $j = 1, \ldots, M$, is equal to 1 if $i = j$ and equal to 0 otherwise.

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = \big(\mathbf{D}_1 \; \mathbf{D}_2 \; \ldots \; \mathbf{D}_M\big)$$

The M-vector $\boldsymbol{\eta}$ has typical element v_i, and so it follows that the n-vector $\mathbf{D}\boldsymbol{\eta}$ has element v_i in the row corresponding to observation it. The vector $\boldsymbol{\eta}$ plays the role of a parameter vector, and it is in this sense that the v_i are called fixed effects. They could in fact be random; the essential thing is that they must be independent of the error terms ε_{it}. That is $\mathrm{E}(\boldsymbol{\varepsilon} \mid \mathbf{D}, \mathbf{X}) = 0$. Note that $\mathrm{Cov}(\mathbf{X}, \mathbf{D}) = 0$ is not needed. The Augmented model:

$$\mathbf{y} = \mathbf{Z}\begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{\eta} \end{bmatrix} + \boldsymbol{\varepsilon}, \quad \mathbf{Z} = \begin{bmatrix} \mathbf{X} & \mathbf{D} \end{bmatrix}.$$

Then we get dummy variables least squares estimator

$$\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\eta}} \end{bmatrix} = \big(\mathbf{Z}'\mathbf{Z}\big)^{-1}\mathbf{Z}'\mathbf{y}$$

Define $\mathbf{P}_D = \mathbf{D}\,(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'$ and $\mathbf{M}_D = \mathbf{I} - \mathbf{P}_D$. By FWL, we get

$$\mathbf{M}_D\mathbf{y} = \mathbf{M}_D\mathbf{X}\boldsymbol{\beta} + \mathbf{M}_D\boldsymbol{\varepsilon}$$

OLS on the demeaned variables gives the within (fixed-effects) estimator

$$\hat{\boldsymbol{\beta}}_{FE} = \big(\mathbf{X}'\mathbf{M}_D\mathbf{X}\big)^{-1}\mathbf{X}'\mathbf{M}_D\mathbf{y}, \quad \mathrm{Var}\big(\hat{\boldsymbol{\beta}}\big) = \big(\mathbf{X}'\mathbf{M}_D\mathbf{X}\big)^{-1}\sigma_\varepsilon^2$$

We now calculate $\mathbf{M}_D$.

$$\mathbf{D}'\mathbf{D} = \begin{bmatrix} T & 0 & \cdots & 0 \\ 0 & T & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T \end{bmatrix} \Rightarrow (\mathbf{D}'\mathbf{D})^{-1} = \begin{bmatrix} \frac{1}{T} & 0 & \cdots & 0 \\ 0 & \frac{1}{T} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{T} \end{bmatrix}$$

$$\mathbf{D}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 & \\ 0 & 0 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 1 \end{bmatrix} \begin{bmatrix} y_{11} \\ \vdots \\ y_{1T} \\ \vdots \\ y_{M1} \\ \vdots \\ y_{MT} \end{bmatrix} = \begin{bmatrix} \sum_t y_{1t} \\ \vdots \\ \sum_t y_{Mt} \end{bmatrix} \Rightarrow (\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'\mathbf{y} = \begin{bmatrix} \frac{1}{T}\sum_t y_{1t} \\ \vdots \\ \frac{1}{T}\sum_t y_{Mt} \end{bmatrix}$$

Thus,

$$\mathbf{P}_D\mathbf{y} = \mathbf{D}\left(\mathbf{D}'\mathbf{D}\right)^{-1}\mathbf{D}'\mathbf{y} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{T}\sum_t y_{1t} \\ \frac{1}{T}\sum_t y_{2t} \\ \vdots \\ \frac{1}{T}\sum_t y_{Mt} \end{bmatrix} = \begin{bmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_2 \\ \vdots \\ \bar{y}_M \\ \vdots \\ \bar{y}_M \end{bmatrix}_{mT\times 1}$$

Then

$$\mathbf{M}_D\mathbf{y} = (\mathbf{I} - \mathbf{P}_D)\mathbf{y} = \mathbf{y} - \mathbf{P}_D\mathbf{y} = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1T} \\ y_{21} \\ \vdots \\ y_{2T} \\ \vdots \\ y_{M1} \\ \vdots \\ y_{MT} \end{bmatrix} - \begin{bmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_2 \\ \vdots \\ \bar{y}_M \\ \vdots \\ \bar{y}_M \end{bmatrix} = \begin{bmatrix} y_{11} - \bar{y}_1 \\ \vdots \\ y_{1T} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{2T} - \bar{y}_2 \\ \vdots \\ y_{M1} - \bar{y}_M \\ \vdots \\ y_{MT} - \bar{y}_M \end{bmatrix}.$$

So we call $\mathbf{M}_D$ Deviations from Group Means. For this reason, $\hat{\boldsymbol{\beta}}_{FE}$ is often called the within-groups estimator. Because $\mathbf{X}$ and $\mathbf{D}$ are exogenous, this estimator is unbiased. Moreover, since the conditions of the Gauss–Markov theorem are satisfied, we can conclude that the fixed-effects estimator is BLUE.

5.5.3 Random-Effects Estimation

For model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad \mathbf{u} = \mathbf{v} + \boldsymbol{\varepsilon},$$

the restrictions imposed for random-effects estimation require that $\mathrm{E}(v_i \mid \mathbf{X}_{it}) = 0$. It follows that

$$\mathbb{E}(u_{it} \mid \mathbf{X}) = \mathbb{E}(v_i + \varepsilon_{it} \mid \mathbf{X}) = 0,$$

so OLS is unbiased and consistent $\hat{\boldsymbol{\beta}}_{\text{OLS}} \xrightarrow{p} \boldsymbol{\beta}$. We also assume the error covariance is

$$\boldsymbol{\Omega} = \begin{bmatrix} \boldsymbol{\Sigma} & & \\ & \ddots & \\ & & \boldsymbol{\Sigma} \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_v^2 + \sigma_\varepsilon^2 & \sigma_v^2 & \cdots & \sigma_v^2 \\ \sigma_v^2 & \sigma_v^2 + \sigma_\varepsilon^2 & \cdots & \sigma_v^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_v^2 & \sigma_v^2 & \cdots & \sigma_v^2 + \sigma_\varepsilon^2 \end{bmatrix}.$$

Then OLS is inefficient but it gives what we need for GLS estimation.
To obtain GLS estimates of $\boldsymbol{\beta}$, we would need to know the values of σ_ε^2 and σ_v^2.

For $\sigma_\varepsilon^2 + \sigma_v^2$, we can use

$$s_u^2 = \frac{1}{n-k}\sum_t \sum_i \hat{u}_{it}^2 \to \sigma_u^2$$

Then we get $\hat{\sigma}_u^2$. For σ_v^2,

$$\mathbf{P}_D\mathbf{y} = \mathbf{P}_D\mathbf{X}\boldsymbol{\beta} + \text{RES} = \mathbf{P}_D\mathbf{X}\boldsymbol{\beta} + \underbrace{\mathbf{P}_D\mathbf{v}}_{\mathbf{v}} + \underbrace{\mathbf{P}_D\boldsymbol{\varepsilon}}_{\text{very small}}, \quad \mathbf{P}_D\mathbf{y} \to \hat{\sigma}_v^2$$

Then we get $\hat{\sigma}_v^2$. By using these two estimators, we can get the GLS estimate

$$\hat{\boldsymbol{\beta}}_{\text{GLS}} = \left(\mathbf{X}'\hat{\boldsymbol{\Omega}}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\hat{\boldsymbol{\Omega}}^{-1}\mathbf{y}, \quad \text{with} \quad \text{Var}(\hat{\boldsymbol{\beta}}_{\text{GLS}}) = \left(\mathbf{X}^\top\hat{\boldsymbol{\Omega}}^{-1}\mathbf{X}\right)^{-1}.$$

5.6 Where are we?

Before we dive into the last chapter, let's review what we have done so far.
Our journey begin with revision of Linear algebra because we need to deal with multiple variables regression afterwards. Then we use Method of Moments to derive the OLS estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

Always remember there are three assumptions:

- Linearity in parameters: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad m > K$
- Exogeneity: $E(u_t \mid x_t) = 0$ $\big(E(x_{it}u_t) = 0\ \forall i = 1, \ldots, K\big)$. Note that there is a strong version $E(\mathbf{u} \mid \mathbf{X}) = 0$.
- Full rank of $\mathbf{X}$: $\mathbf{X}^\top\mathbf{X}$ is invertible (no perfect multicollinearity)

After that, we look at the geometry of OLS estimation. It turns out that any $\mathbf{y}$ decomposes as $\mathbf{y} = \mathbf{P}_\mathbf{X}\mathbf{y} + \mathbf{M}_\mathbf{X}\mathbf{y} = \hat{\mathbf{y}} + \hat{\mathbf{u}}$, yielding the Pythagorean identity $TSS = ESS + SSR$. By using the projection matrices, we can prove the FWL theorem, which can help us estimate any sub-coefficient. Beyond that, FWL can also help us measure the goodness of fit and leverage.
Moreover, we move into statistical properties of OLS (we implicitly assume the above three assumptions hold to give us the OLS estimator):

- Under $E(\mathbf{u} \mid \mathbf{X}) = 0$, the OLS estimator is unbiased.
- Under $E(u_t \mid \mathbf{x}_t) = 0$ and $\text{plim}_{m\to\infty} \frac{1}{m}\mathbf{X}^\top\mathbf{X} = \mathbf{S}_{\mathbf{X}^\top\mathbf{X}}$, the OLS estimator is consistent.
- Define the covariance matrix $\text{Var}(\hat{\boldsymbol{\beta}}) = E\Big[(\hat{\boldsymbol{\beta}} - E(\hat{\boldsymbol{\beta}})(\hat{\boldsymbol{\beta}} - E(\hat{\boldsymbol{\beta}})'\Big]$. Under $\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$ and spherical disturbances $\text{Var}(\mathbf{u}) = E(\mathbf{u}\,\mathbf{u}') = \sigma^2\mathbf{I}_M$, $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. We use $s^2 = \frac{M}{M-k}\hat{\sigma}^2 = \frac{1}{M-k}\sum_t \hat{u}_t^2$ to obtain an unbiased estimate of $\text{Var}(\hat{\boldsymbol{\beta}})$, $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = s^2(\mathbf{X}^\top\mathbf{X})^{-1}$.
- Under $\mathbb{E}(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$, and $\mathbb{E}(\mathbf{u}\mathbf{u}^\top) = \sigma^2\mathbf{I}$, Gauss–Markov theorem holds: the OLS estimator is BLUE.

There are two types of misspecification: overspecification and underspecification. We conclude that underspecification is a much more severe problem than overspecification.
If we wanna test a single restriction, we assume $\mathbf{u} \sim \mathcal{N}\big(\mathbf{0}, \sigma^2\mathbf{I}\big)$ and use t-test. If we wanna test several restrictions, we assume $\mathbf{u} \sim \mathcal{N}\big(\mathbf{0}, \sigma^2\mathbf{I}\big)$ and use F-test.
The spherical disturbances is too strong. In general, we have $\mathbb{E}(\mathbf{u}\mathbf{u}^\top) = \boldsymbol{\Omega}$. It turns out that OLS estimator is no longer efficient. We use a new estimator called GLS: $\hat{\boldsymbol{\beta}}_{\text{GLS}} = \left(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}$, with $\text{Var}(\hat{\boldsymbol{\beta}}_{\text{GLS}}) = \left(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}\right)^{-1}$. GLS estimator is more efficient. We can use $\hat{\boldsymbol{\gamma}}$ to estimate $\boldsymbol{\Omega}$.
For more complicated data, we need to use panel data. There are two scenarios:

- If the individual effect is exogenous, $\text{E}(v_i \mid \mathbf{X}_{it}) = 0$, GLS equals the random-effects estimator and is efficient $\hat{\boldsymbol{\beta}}_{\text{GLS}} = \hat{\boldsymbol{\beta}}_{\text{RE}}$.
- If that exogeneity fails $\text{E}(v_i \mid \mathbf{X}_{it}) \neq 0$, RE/GLS is biased; switch to fixed-effects or other consistent methods.

Chapter 6

Instrumental Variables

Instrumental Variables are used to solve Endogeneity problem: when $\mathrm{E}(u_t \mid X_t) \neq 0$, the OLS estimator on $y_t = \mathbf{1}_M \beta_1 + \underbrace{X_t}_{M\times 1} \underbrace{\beta_2}_{1\times 1} + u_t$ is biased and inconsistent.

$$\begin{cases} \mathbf{y} = \mathbf{1}_M \beta_1 + \mathbf{X}\beta_2 + \mathbf{u}, \quad \mathrm{E}(u_t \mid X_t) \neq 0 \quad \text{(Structural equation)} \\ \mathbf{X} = \mathbf{W}\Pi + \mathbf{v}, \quad \mathrm{E}(u_t \mid v_t) \neq 0 \quad \text{(Reduced-form equation)} \end{cases}$$

where $\mathbf{W}_{M\times 1}$ is a candidate instrument. There are two validity conditions for $\mathbf{W}$:

(i) Exogeneity: $\mathrm{E}(v_t \mid W_t) = 0$ (exogeneity in reduced form) and $\mathrm{E}(u_t \mid W_t) = 0$ (exogeneity w.r.t u_t).

(ii) Relevance: $\Pi \neq 0$. If $\Pi \simeq 0$, $\mathbf{W}$ is a weak instrument.

If $\mathbf{W}$ is exogenous and relevant, it is a valid instrument variable.

6.1 Instrumental Variables Estimation

6.1.1 Two-Stage Least Square

Let's consider the simple case where $\mathbf{X}$ is an $M \times 1$ vector.

$$\begin{cases} \mathbf{y} = \mathbf{1}_M \beta_1 + \mathbf{X}\beta_2 + \mathbf{u} = \mathbf{1}_M \beta_1 + \mathbf{x}\beta_2 + \mathbf{W}\delta + \mathbf{u}, \quad \delta = 0 \\ \mathbf{X} = \mathbf{W}\Pi + \mathbf{v}, \quad \mathrm{E}(v_t \mid W_t) = 0 \end{cases}$$

where $\delta = 0$ is exclusion restriction to make sure $\mathbf{W}$ affect $\mathbf{y}$ only through $\mathbf{X}$.
Two-Stage Least Squares (TSLW):

1. First-stage: Estimate $\hat{\Pi} = (\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{X}$ with $\hat{\Pi} \xrightarrow{p} \Pi$ and get the fitted values

$$\hat{\mathbf{X}} = \mathbf{W}\hat{\Pi} = \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'\mathbf{X} = \mathbf{P}_{\mathbf{W}}\mathbf{X} \tag{6.1}$$

   Because $\mathrm{E}(v_t \mid W_t) = 0$, we also have $\mathrm{E}(v_t \mid \hat{x}_t) = 0$.
   Rule of thumb: an instrument variable is weak if the F-test on the reduced form is less than 10, namely $F < 10$.

2. Second-stage: Run OLS of $\mathbf{y} = \mathbf{1}_M \beta_1 + \hat{\mathbf{X}}\beta_2 + \mathbf{u}$. We get the Two-stage least-squares estimator:

$$\hat{\beta}_2 = \left(\hat{\mathbf{X}}'\mathbf{M}_1\hat{\mathbf{X}}\right)^{-1}\hat{\mathbf{X}}'\mathbf{M}_1\mathbf{y} \xrightarrow{p} \beta_2 \tag{6.2}$$

What if we have more than one IVs? Suppose we have 1 regressor. Consider the model

$$\begin{cases} \mathbf{y} = \mathbf{1}_M \beta_1 + \underbrace{\mathbf{X}}_{M\times 1} \beta_2 + \mathbf{u} \\ \mathbf{X} = \underbrace{\mathbf{W}}_{M\times \ell} \mathbf{\Pi} + \mathbf{v}, \quad \ell \in \mathbb{Z}_+ \end{cases}$$

- $\ell < 1 \Rightarrow$ under-identified.
- $\ell = 1 \Rightarrow$ just-identified.
- $\ell > 1 \Rightarrow$ over-identified.

For $\ell = 2$, there are three ways to do first-stage estimation:

$$\begin{cases} (A) & \mathbf{X} = \mathbf{W}_1\Pi_1 + \mathbf{W}_2\Pi_2 + \mathbf{v} \Rightarrow \hat{\mathbf{X}}_A \\ (B) & \mathbf{X} = \mathbf{W}_1\Pi_1 + 0 \qquad + \mathbf{v} \Rightarrow \hat{\mathbf{X}}_B \\ (C) & \mathbf{X} = 0 \qquad + \mathbf{W}_2\Pi_2 + \mathbf{v} \Rightarrow \hat{\mathbf{X}}_C \end{cases}$$

6.1.2 The Generalized IV Estimator

For $\underbrace{\mathbf{W}}_{M\times\ell}$ and $\underbrace{\mathbf{X}}_{M\times K}$, where $M \gg K$, $M \gg \ell$, and $\operatorname{rank}(\mathbf{W}) = \ell$,

$$\begin{cases} \mathbf{y} = \underbrace{\mathbf{X}}_{M\times K}\boldsymbol{\beta} + \mathbf{u} \\ \underbrace{\mathbf{X}}_{M\times K} = \underbrace{\mathbf{W}}_{M\times\ell}\underbrace{\boldsymbol{\Pi}}_{\ell\times K} + \underbrace{\mathbf{v}}_{M\times K} \Rightarrow \underbrace{\mathbf{X}_k}_{M\times 1} = \underbrace{\mathbf{W}}_{M\times\ell}\underbrace{\boldsymbol{\Pi}_k}_{\ell\times 1} + \underbrace{\mathbf{v}_k}_{M\times 1}, \quad k = 1,\ldots,K \end{cases}$$

- $\ell < K \Rightarrow$ under-identified.
- $\ell = K \Rightarrow$ identified.
  In this case, exogeneity assumption $\mathrm{E}(\mathbf{u} \mid \mathbf{W}) = 0$ gives $\mathbf{W}'\mathbf{u} = \mathbf{0}$. We also have $\mathbf{u} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$. Then the Moment condition is $\mathbf{W}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$. There are K unknowns and K equations. Solving the moment system yields the simple IV estimator:

  $$\mathbf{W}'\mathbf{y} - \mathbf{W}'\mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{0} \Rightarrow \hat{\boldsymbol{\beta}}_{\mathrm{IV}} = (\mathbf{W}'\mathbf{X})^{-1}\mathbf{W}'\mathbf{y}$$

  Using the projection matrix $\mathbf{P}_{\mathbf{W}}$, we get $\hat{\mathbf{X}} = \mathbf{P}_{\mathbf{W}}\mathbf{X}$. Then the estimator can be written in two-stage form (generalized IV / 2SLS):

  $$\hat{\boldsymbol{\beta}}_{\mathrm{GIV}} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y} = \left(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y}$$

- $\ell > K \Rightarrow$ over-identified.
  In this case, exogeneity assumption $\mathrm{E}(\mathbf{u} \mid \mathbf{W}) = 0$ gives $\mathbf{W}'\mathbf{u} = \mathbf{0}$. We also have $\mathbf{u} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$. Then the Moment condition is $\mathbf{W}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$. There are K unknowns and ℓ equations but $\ell > K$.

  $$\begin{cases} \mathbf{W}_1\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}, \\ \mathbf{W}_2\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}, \\ \vdots \\ \mathbf{W}_\ell\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}. \end{cases}$$

  We can always treat an overidentified model as if it were just identified by choosing exactly K linear combinations of the ℓ columns of $\mathbf{W}$. Formally, we seek an $\ell \times K$ matrix $\mathbf{J}$ such that the $M \times K$ matrix $\mathbf{WJ}$ is a valid instrument matrix and such that the use of $\mathbf{J}$ minimizes the asymptotic covariance matrix of the estimator in the class of IV estimators. Thus,

  $$\underset{K\times\ell}{\mathbf{J}^{\mathsf{T}}}\,\underset{\ell\times M}{\mathbf{W}^{\mathsf{T}}}\,\underset{M\times 1}{\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right)} = \mathbf{0}.$$

  It turns out that

  $$\mathbf{X}'\mathbf{P}_{\mathbf{W}} = \mathbf{X}'\mathbf{W}\,(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$$

  is the best choice. Then

  $$\mathbf{J}'\mathbf{W}' = \mathbf{X}'\mathbf{P}_{\mathbf{W}} = \mathbf{X}'\mathbf{W}\,(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$$

  Then

  $$\mathbf{J}'\mathbf{W}'\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = 0 \Rightarrow \mathbf{X}'\mathbf{P}_{\mathbf{W}}\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y} - \mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\,\hat{\boldsymbol{\beta}} = 0$$

Solving for β yields the Generalised IV or 2-stage least-squares estimator:

$$\hat{\boldsymbol{\beta}}_{\text{GIV}} = \left(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y} = \left(\hat{\mathbf{X}}'\hat{\mathbf{X}}\right)^{-1}\hat{\mathbf{X}}'\mathbf{y}$$

It is consistent. Indeed,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y} = (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}(\mathbf{X}\boldsymbol{\beta}+\mathbf{u}) = (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\,\boldsymbol{\beta}+(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{u}$$

$$= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{u} \xrightarrow{p} \boldsymbol{\beta}$$

because $\frac{1}{m}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X} \xrightarrow{p} S_{wx}$ and $\frac{1}{m}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{u} \xrightarrow{p} 0$.
We can also calculate the variance of the estimator. We assume $\mathrm{E}(\mathbf{u}\mathbf{u}') = \sigma^2\mathbf{I}$ to make life easier. Then,

$$\mathrm{Var}\big(\hat{\boldsymbol{\beta}}_{\text{GIV}}\big) = \mathrm{Var}\Big(\boldsymbol{\beta} + (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{u}\Big) = \mathrm{E}\Big[(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{u}\,\mathbf{u}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\,(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\Big]$$

$$= (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\,\mathrm{E}(\mathbf{u}\mathbf{u}')\,\mathbf{P}_{\mathbf{W}}\mathbf{X}\,(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\,(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}$$

$$= \sigma^2(\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1} = \sigma^2(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}$$

6.2 Hausman test

In many cases, we do not know whether we actually need to use instrumental variables. In this section, we discuss a simple procedure that can be used to decide whether we need IV or not. We call it Durbin–Wu–Hausman tests or DWH tests.
The null and alternative hypotheses for the DWH test can be expressed as

$$\begin{cases} H_0: \; y_t = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, & u_t \sim \mathrm{IID}(0,\sigma^2), \quad \mathrm{E}(u \mid X) = 0 \\ H_1: \; y_t = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, & u_t \sim \mathrm{IID}(0,\sigma^2), \quad \mathrm{E}(u \mid \mathbf{W}) = 0 \text{ but } \mathrm{E}(u \mid X) \neq 0 \end{cases}$$

Under H_1, the IV estimator $\hat{\boldsymbol{\beta}}_{\text{GIV}}$ is consistent, but the OLS estimator $\hat{\boldsymbol{\beta}}_{\text{OLS}}$ is not. Under H_0, both are consistent. The idea of the DWH test is to check whether the difference $\hat{\boldsymbol{\beta}}_{\text{IV}} - \hat{\boldsymbol{\beta}}_{\text{OLS}}$ is significantly different from zero in the available sample. This difference can be written as

$$\big(\hat{\boldsymbol{\beta}}_{\text{GIV}} - \hat{\boldsymbol{\beta}}_{\text{OLS}}\big) = (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$= (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\big[\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{y} - \mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\big]$$

$$= (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\big[\mathbf{I} - \mathbf{X}\,(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\big]\mathbf{y} = (\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{y}.$$

Therefore, testing whether $\hat{\boldsymbol{\beta}}_{\text{IV}} - \hat{\boldsymbol{\beta}}_{\text{OLS}}$ is significantly different from zero is equivalent to testing whether the vector $\mathbf{X}^\top\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{y}$ is significantly different from zero.
Under H_0, the preferred estimation technique is OLS, and the OLS residuals are given by the vector $\mathbf{M}_{\mathbf{X}}\mathbf{y}$. Therefore, we wish to test whether the k columns of the matrix $\mathbf{P}_{\mathbf{W}}\mathbf{X}$ are orthogonal to this vector of residuals. Let us partition the matrix $\mathbf{X}$, so that $\mathbf{X} = [\,\mathbf{Z}\;\mathbf{Y}\,]$, where the k_1 columns of $\mathbf{Z}$ are included in the matrix of instruments $\mathbf{W}$, and the $k_2 = k - k_1$ columns of $\mathbf{Y}$ are treated as potentially endogenous. By construction, OLS residuals are orthogonal to all the columns of $\mathbf{X}$, in particular to those of $\mathbf{Z}$. The test is thus concerned only with the k_2 elements of $\mathbf{Y}^\top\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{y}$, which will not in general be identically zero, but should not differ from it significantly under H_0.
The easiest way to test whether $\mathbf{Y}^\top\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{y}$ is significantly different from zero is to use an F test for the k_2 restrictions $\boldsymbol{\delta} = \mathbf{0}$ in the OLS regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{P}_{\mathbf{W}}\mathbf{Y}\boldsymbol{\delta} + \mathbf{u}.$$

The OLS estimates of $\boldsymbol{\delta}$ are, by the FWL Theorem, the same as those from the FWL regression of $\mathbf{M}_{\mathbf{X}}\mathbf{y}$ on $\mathbf{M}_{\mathbf{X}}\mathbf{P}_{\mathbf{W}}\mathbf{Y}$, that is,

$$\hat{\boldsymbol{\delta}} = \left(\mathbf{Y}^\top\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{P}_{\mathbf{W}}\mathbf{Y}\right)^{-1}\mathbf{Y}^\top\mathbf{P}_{\mathbf{W}}\mathbf{M}_{\mathbf{X}}\mathbf{y}.$$